What Ten Industrial Surveys Identify in a Cone-Crusher Population Balance Model: Bayesian Calibration with Structural Discrepancy Narrows the Engine's Power Band but Gives No Accuracy Gain over a Setting-Only Line

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***Abstract*. A point fit of a cone-crusher model to a few industrial surveys does not say which fitted scalars the surveys determine or how well a new survey is predicted. We calibrate the eight fitted scalars of a Whiten classification and breakage engine of a Metso HP500 cone crusher as a Bayesian posterior on the ten surveys of Rocha et al., with a Kennedy and O'Hagan structural discrepancy linear in the closed-side setting and the feed size, integrated out exactly and sampled with differential-evolution moves. Strict leave-one-survey-out validation refits every quantity per fold; eleven other methods run on the same folds. Held out, the predictive mean reaches 12.00 % (95 % bootstrap interval 7.7 to 15.9) on throughput and 14.31 % (9.0 to 19.5) on power; against the setting-only line (12.78 %) the paired difference on power is 1.54 points (−1.11 to 4.10), no gain. The 80 % bands cover 8 and 9 of the ten held-out surveys and are 43.9 % narrower on power than the residual band of the refit engine, about as wide as the line's band. We prove that the engine's throughput lies in the span of the discrepancy, so the surveys narrow the capacity base only through the scale of the discrepancy prior (contraction 0.294 to 0.638 as that scale is halved or doubled), that the predictive mean is a shrunk ridge regression, and that the work index and the power slope enter only as a product. The surveys identify the noise and the setting dependence; the population-balance scalars are weakly identified.**

***Index Terms*. Cone crusher, population balance, Bayesian calibration, model discrepancy, identifiability, leave-one-out validation, differential-evolution MCMC, comminution.**

I. Introduction

THE electricity of a copper concentrator goes largely to crushing and grinding, and in Chile in 2024 the concentrators took 55.7 % of the electricity of copper mining [1]. The setting of a crusher decides how much work the grinding downstream must do, so a model that predicts the throughput, the power and the product size of a cone crusher from its closed-side setting and its feed is a planning tool of direct economic weight. Such models are written in the tradition of Whiten [2]: a classification function decides which particles are nipped and broken, a breakage function distributes the progeny, and a mass balance gives the product. The classification constants, the breakage law, the capacity and the power form of an engine of this kind are fitted to survey data, and the fitted values are then used as if they were known.

Industrial surveys of a crusher are few and expensive. Rocha et al. [8] report ten surveys of an HP500 secondary cone crusher on itabirite iron ore with four numbers each that are tabulated (the closed-side setting, the feed size f_{80}, the throughput and the power, the last estimated from electric current) and a product size distribution that appears only as a figure. They fitted a model of the machine on the same ten surveys. A point fit to ten rows of two measured outputs leaves two questions open. Which of the fitted scalars do the surveys determine, and which does the fit leave where it started? And how well, and with what uncertainty, is an unseen survey predicted? The published crusher models we found are calibrated by least squares or by iterative tuning, and they give point estimates. Duarte et al. [9] do validate a least-squares calibration of the product size distribution on an industrial data set left out of the fit, but no work we found gives a posterior of the fitted scalars, a predictive interval, or a survey-by-survey held-out evaluation of the throughput and the power (Section II). The statistical literature on computer-model calibration gives the tools to answer both questions, a posterior over the parameters and a discrepancy term that keeps the model error out of the parameters [27], [28], [30], together with an analysis of when the two can be told apart [31], [32], [33], but we found no application of it to a crusher population balance.

We propose CrusherBayes, a Bayesian calibration of the engine with an explicit structural discrepancy. The engine is

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Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChancaDEM.

run as its own simulator, the eight fitted scalars receive weakly informative priors centred on the engine's shipped values, most of which are fitted numbers of Rocha et al., and a discrepancy linear in the standardized setting and feed size is added to each measured output in the form of Kennedy and O'Hagan [27]. Because the discrepancy and the noise are Gaussian given the scalars, the discrepancy is integrated out exactly and the sampler explores ten dimensions. The method gives a posterior per scalar, a predictive distribution per held-out survey and a table of prior against posterior. Our contributions are:

1) the method: a Bayesian calibration with exact marginalization of a structural discrepancy for a cone-crusher population balance engine, sampled with differential-evolution moves chosen by an autocorrelation criterion;
2) a theory of what the surveys can identify, which states for this engine and this discrepancy the exact form of the confounding of calibration parameters and discrepancy that the calibration literature treats in general [30], [31], [33]: the engine's throughput lies exactly in the column space of the discrepancy design, the posterior predictive mean is a ridge regression shrunk toward the engine's best affine approximation, and the work index and the power slope enter only as a product, with a linearized posterior that, compared with the sampler on the same scale, matches the sampled contraction within 0.024 on thirteen of the fourteen engine and discrepancy parameters;
3) a strict leave-one-survey-out comparison, with bootstrap intervals and paired differences, against eleven methods run on the same folds: the setting-only line, the free-form linear fit, the engine with its capacity base and its power form refit, the same with a bounded residual, Morrell's size-specific energy with and without a no-load term, the Bond draw, the engine at its shipped constants, the constant mean, and the published throughput model of Rocha et al. as published and with its fitting parameter refit in each fold;
4) an ablation of the discrepancy, the engine scalars, the population-balance scalars, the prior widths, the discrepancy scale and the sampler, with three seeds of the full method.

The main result is partly negative. Held out, CrusherBayes is as accurate as the refit engine and the published throughput model of Rocha et al. on throughput and no more accurate than a line in the closed-side setting on power; its 80 % bands cover 8 and 9 of the ten surveys, a coverage that ten surveys cannot tell apart from that of a slightly miscalibrated band, and they are narrower than the residual band of the refit engine. The surveys identify the noise and the dependence of both outputs on the setting. They narrow the capacity base only through the scale of the discrepancy prior: the engine's throughput lies in the span of the discrepancy (Proposition 2), so the shape of the throughput data cannot determine it, and its ratio of posterior to prior standard deviation moves from 0.294 to 0.454 to 0.638 as that scale is halved, kept and doubled (Section VIII), across the threshold of 0.5 that we read as identified. The classification and breakage scalars of the population balance are weakly identified: their posterior standard deviation stays above nine tenths of the prior's, and the posterior means of the two classification multipliers move by about a third of a prior standard deviation.

Section II places the method among the published approaches; Section III states the problem; Section IV gives the method; Section V proves what it can identify; Sections VI to VIII give the design, the results and the ablation, and Sections IX to XII discuss them.

II. Related Work

A. Population balance models of cone crushers

Whiten [2] modelled a crusher as a classification and breakage process solved as a mass balance; Napier-Munn et al. [3] give the breakage descriptors (t_{10} and the ore's $A \times b$) and the classification relations that such a model is calibrated with, and Andersen and Napier-Munn [4] gave a parameterization of the classification constants and a power prediction for cone crushers. Evertsson modelled the flow of material in a cone crusher [6] and its capacity and performance [7]. Bond's third theory [5] gives the specific energy of a size reduction from the feed and product P_{80}. Rocha et al. [8] fitted an Andersen–Whiten model of the HP500 to the ten surveys used here. Their classification constants K_1 and K_2 and their breakage index t_{10} are linear in the setting and the feed size, with $K_3 = 2.3$, a value they keep constant after earlier work (their p. 11). Their maximum throughput is $Q_{\max} = Q_T\, \alpha\beta\gamma\delta\epsilon\, \lambda$ (their Eq. 10), the capacity $Q_T = 647$ t/h of a base case simulated in Metso's Bruno at a setting of 38 mm, a granite of work index 16 kWh/t and a feed f_{80} of 100 mm, multiplied by correction factors for the bulk density, the work index, the feed size, the moisture and the setting and by a fitting parameter $\lambda_{\mathrm{R}} = 0.84$ (their Section 2.2 and Table 4). Their power is $P_c = \zeta P_d + P_n$ after Andersen and Napier-Munn, with P_d the drop-weight energy of the feed classified for breakage times the throughput (their Eqs. 7 and 8), $\zeta = 1.30$ fitted to the surveys and $P_n = 110$ kW set to a third of the installed power (their Table 5 and Appendix C); they report only fair agreement on power and relate it to the measurement of power on crushers operating in parallel. Their fitting parameter λ_{R} was set on survey 10 alone, while the exponents of the correction factors come from simulations in Bruno and from crusher manuals (their Section 4.3 and Appendix B), so their throughput model takes one number from these surveys. The engine of this work takes their classification coefficients, their t_{10} relation, K_3, ζ and P_n as numbers, but neither their throughput model nor their P_d: its capacity is a reduced Evertsson form and its P_d a Bond draw (Section IV), so the engine at its shipped constants is not the model of Rocha et al. We take those numbers as the nominal values of our priors. Duarte et al. [9] calibrate a Whiten cone-crusher model with industrial data from two iron-ore mines, with every entry of the breakage matrix as a calibration parameter fitted by least squares to the product size distribution, and they validate it out of sample: calibrated on two data sets of one mine and validated on the third, which the fit did not use (their Scenario 2, Section 5.2), and calibrated on all the data sets of a mine and used to predict the

product at other settings (their Scenario 3). Their target is the product size distribution, not the throughput or the power, and their result is a point estimate scored by a sum of squared errors, with no interval. Lindqvist and Li [10] couple the energy and the breakage dependency of a cone-crusher model. Bengtsson [11] predicts the size distribution from a first-order differential equation for the coefficient of variance as a function of the compression length, combined with a bimodal Weibull distribution, and models the power consumption with the same Weibull analysis; it describes compression breakage directly rather than through a population balance, and it is not calibrated as a posterior. Atta et al. [12] extend a control-oriented cone-crusher model with energy consumption and validate it on laboratory and field data, and Liu et al. [13] compute the energy of a cone crusher as the product of the compressive pressure and the displacement on the liner. Yamashita et al. [14] review the modelling and control of cone crushers. These works give the model structure and the point estimates, and one of them, Duarte et al., validates a calibration of the product size on industrial data held out of the fit; none of them reports a posterior of the fitted scalars, a predictive interval, or a held-out evaluation of the throughput and the power of a calibrated crusher survey by survey.

B. Energy and power of crushing

Morrell [15] predicts the specific energy of crushing, high-pressure grinding and tumbling mill circuits from a size-specific energy; the guideline of the Global Mining Guidelines Group [16] gives its crusher equations (B4 and B5) with the crusher work index M_{ic} of the SMC test, and defines the net power as the motor power less the no-load power. Both Bond and Morrell need a work index of the ore; the Morrell index of the Minas Rio itabirite is not published, so here it is a fitted scalar. The power form of Andersen and Napier-Munn [4], a slope times a net power plus a no-load power, is the power model of the engine, with a Bond draw as its net power where Rocha et al. [8] use the drop-weight energy. The three forms differ in what they assume about the measured power: Bond's draw is a net comminution power with no idle term, Morrell's guideline subtracts a no-load power that the surveys do not report, and the Andersen and Napier-Munn form fits both a slope and a no-load term to the motor power. On ten surveys each of them is therefore a fitted relation, and Section VII runs all three on the same folds.

C. Discrete-element and learned models

Discrete-element simulation with breakage is the detailed alternative to the population balance. Cleary and Sinnott [17] simulate particle flows and breakage in compression crushers, Quist and Evertsson [18] model and simulate a cone crusher with the discrete-element method, Johansson et al. [19] evaluate the performance of a cone crusher with simulations and laboratory experiments, André and Tavares [20] simulate a laboratory cone crusher with polyhedral particles, and Moncada et al. [21] predict crushing power in cone crushers with the discrete-element method. Metso report a fast-breakage discrete-element cone model developed over about a decade and regression equations fitted to it [22]. These simulations need a liner geometry and hours of computation per second of operation, and several of them are validated on laboratory machines; the HP500 liner geometry is not public. On the learned side, Koh et al. [23] use a deep network as a surrogate of the phenomenological models of a comminution circuit, Salehzadeh Gharaei et al. [24] simulate the power of an iron-ore cone crusher plant with a multi-method machine-learning framework, and Chen et al. [25] and Gupta and Mishra [26] solve or learn breakage population balances with physics-informed or multi-head networks. A learned model of ten surveys is a risk of overfitting; here the free-form linear fit and the setting-only line stand for this family.

D. Calibration of computer models with discrepancy

Kennedy and O'Hagan [27] calibrate the parameters of a computer model against field data by writing the data as the simulator at the true parameters plus a model discrepancy, itself a Gaussian process, plus measurement error. Higdon et al. [28] combine field data and simulations for calibration and prediction with Gaussian-process models of the unknown functions, for a simulator that cannot be run at every parameter value, and Bayarri et al. [29] give a framework for validating a computer model against reality. Brynjarsdóttir and O'Hagan [30] show that an analysis that ignores the discrepancy gives biased and over-confident parameter estimates, and that the confounding of the discrepancy with the calibration parameters is resolved only with meaningful priors. We use the low-order form of this construction: the discrepancy is linear in two standardized regressors, with a Gaussian prior on its three coefficients per output, which makes the marginal likelihood exact. Section V gives the exact form that this confounding takes for the linear discrepancy and this engine.

The confounding has its own literature. Arendt, Apley and Chen [31] show with several examples that the effects of the calibration parameters and of the discrepancy cannot always be told apart, explain the mechanisms, and find that identifiability is reached under mild assumptions in some systems and is virtually impossible in others; in a companion paper they gain identifiability from several responses that share the calibration parameters. Because the Gaussian-process calibration of Kennedy and O'Hagan can give an unreasonable estimate for an imperfect computer model, Tuo and Wu [32] take the calibration parameter to be the minimizer of the L_2 distance between the physical response and the model, and prove, for physical data with random error, that their L_2 calibration is semiparametric efficient while ordinary least squares is consistent but not efficient. Plumlee [33] notes that the parameter of the predominant Bayesian calibration is left undefined, so its posterior is broader than it should be, and proposes a prior on the discrepancy that is orthogonal to the gradient of the computer model in the parameters. That prior removes by construction the confounding of Proposition 2 below, where the gradient of the throughput in $\ln Q_{\mathrm{ref}}$ lies in the span of the discrepancy. Our propositions do not add a general result to these works: they are exact instances, for a discrepancy linear in two regressors and for this engine, of the confounding

they describe, and they say which scalars it reaches. On the side of comminution models, Bhonsale, Stokbroekx and Van Impe [34] assess the structural and practical identifiability of the breakage parameters of a population balance model of an air jet mill by the profile likelihood, on simulated data, and find the model practically non-identifiable when the size distribution is measured only at the end of milling and identifiable with measurements in time. Their analysis is of a point estimate and of the information in a size measurement; ours is of a posterior with a discrepancy when the size is not measured at all.

E. Samplers and predictive evaluation

The affine-invariant ensemble sampler of Goodman and Weare [35], implemented in `emcee` [36], moves a walker along the line through another walker (the stretch move). ter Braak [37] proposed differential-evolution Markov chain Monte Carlo, whose jumps are built from differences between chains, and ter Braak and Vrugt [38] added a snooker updater so that fewer chains suffice. Both families are gradient free, which matters here because the engine solves a linear system per evaluation in plain Python. For the evaluation of probabilistic predictions, Gneiting et al. [39] assess probabilistic forecasts by their calibration and their sharpness, and Gneiting and Raftery [40] study strictly proper scoring rules for prediction and estimation; we score the coverage, the width and the log predictive density of each held-out prediction in that frame. Yao et al. [41] average Bayesian predictive distributions by stacking, a combination we do not use because ten surveys leave too few degrees of freedom for stable weights. Our intervals on the scores use the bootstrap of Efron [42] and the score interval of Wilson [43] for a binomial proportion.

Searches in Crossref and on the open web for a Bayesian calibration with discrepancy of a crusher or mill population balance returned general methodology and no application to a cone crusher; Scopus and Web of Science were not searched, so the search is not exhaustive. Among the works above, CrusherBayes takes the structure of Whiten and the point estimates of Rocha et al., the inference of Kennedy and O'Hagan with the low-order discrepancy, and the samplers of ter Braak. What it adds for a crusher is a posterior over the fitted scalars with a structural discrepancy, a predictive distribution of the throughput and the power of each held-out survey, a strict leave-one-survey-out comparison of both outputs against eleven methods (among them the published throughput model of Rocha et al.), and the exact form that the confounding of scalars and discrepancy takes in this engine. Duarte et al. [9] validate a point calibration of the product size out of sample, and Bhonsale et al. [34] analyse the identifiability of a mill population balance by the profile likelihood; neither gives a posterior, an interval or the throughput and power of an industrial crusher.

III. Problem Formulation

A. Data

The data are the ten surveys of the Metso HP500 secondary cone crusher at an itabirite iron-ore operation of Rocha et al. [8] (Table A1 of that work, CC BY 4.0), transcribed and validated by a survey parser (Table I). The parser reads each value, refuses one outside its bounds and requires the survey numbers 1 to 10 without a gap. The closed-side setting c spans 38 to 55 mm, the feed f_{80} 47 to 140 mm, the throughput 813 to 1639 t/h and the power 187 to 355 kW. Throughput and power are the two targets. The power is not a metered motor power: the source marks every value of its power column as estimated from electric current, and attributes its own poor fit on power to the measurement of power on crushers operating in parallel [8]. The power target therefore carries a measurement error of unknown size and sign per survey, which the noise scale σ_{kW} and the power discrepancy absorb without telling it apart from model error. The product P_{80} is not measured (the product size distributions of the source are figures only), so it is never a target; the engine's P_{80} over the ten surveys spans 21.2 to 36.6 mm and is a model output. Two correlations of the data, computed from the survey table and the engine's documents, shape every result below. The measured power falls as the setting widens (Pearson correlation -0.82), and the measured throughput rises with it (0.81); power and throughput are correlated at -0.85, while the correlation of the measured power with the engine's P_{80} is -0.14.

TABLE I
THE TEN HP500 SURVEYS OF ROCHA ET AL. (TABLE A1, CC BY 4.0): CLOSED-SIDE SETTING, FEED f_{80}, THROUGHPUT AND POWER OF EACH SURVEY, THE POWER ESTIMATED BY THE SOURCE FROM ELECTRIC CURRENT.

Survey	Setting (mm)	f_{80} (mm)	Throughput (t/h)	Power (kW)
1	54	88	1639	209
2	54	84	1267	212
3	48	47	1240	262
4	48	88	1034	349
5	48	62	1124	325
6	40	79	813	355
7	38	140	1050	306
8	44	123	1138	325
9	40	81	820	350
10	55	81	1334	187

B. The calibration problem

Write $\mathbf{x}_i = (c_i, f_{80,i})$ for the operating point of survey i and $y_{t,i}$ for its measured output $t \in \{\mathrm{tph}, \mathrm{kW}\}$. The engine is a deterministic map $f_t(\boldsymbol{\theta}; \mathbf{x})$ from eight scalars $\boldsymbol{\theta}$ and the operating point to the output. The problem has three parts. First, a posterior $\pi(\boldsymbol{\theta} \mid \mathcal{D})$ over the scalars given the surveys $\mathcal{D}$, with a model of the error between the engine and the plant that does not force that error into the scalars. Second, for a survey h not used in the fit, a predictive distribution $p(y_{t,h} \mid \mathcal{D}_{-h})$ whose mean is accurate and whose central intervals cover the measurement at their nominal rate. Third, the identification question: for each scalar, how much the posterior narrows the prior. The assumptions are those of the engine (one breakage zone, a feed rebuilt from its f_{80}, a fixed throw and speed), that the surveys are exchangeable given the operating point, that the two outputs are independent given the scalars, and that the measurement error is Gaussian with one scale per output.

TABLE II
NOTATION USED THROUGHOUT, WITH THE MEANING OF EACH SYMBOL AND ITS UNIT; EVERY SYMBOL IS DEFINED AGAIN AT ITS FIRST USE IN THE TEXT.

Symbol	Meaning	Unit
c, f_{80}	closed-side setting, feed 80 % passing size	mm
P_{80}, F_{80}	product and feed 80 % passing size (engine)	mm
$y_{t,i}$	measured output t of survey i	t/h or kW
$f_t(\boldsymbol{\theta};\mathbf{x})$	engine output t at scalars $\boldsymbol{\theta}$	t/h or kW
Q_{ref}	capacity base	t/h
W_i	Bond work index of the ore blend	kWh/t
ζ, P_n	slope and no-load power of the power form	–, kW
s_{K_1}, s_{K_2}, K_3	classification multipliers and exponent	–
Δt_{10}	shift of the breakage index t_{10}	points
δ_t, $\mathbf{d}_t$	discrepancy of output t and its coefficients	t/h or kW
$\mathbf{x}_i$	operating point $(c_i, f_{80,i})$ of survey i	mm
$\mathbf{z}_i$, $\tilde{\mathbf{z}}_h$	standardized (c, f_{80}); design row $(1, \mathbf{z}_h)$	–
$\mathbf{Z}$	discrepancy design, rows $(1, \mathbf{z}_i)$	–
σ_t, τ_t	noise scale and discrepancy prior scale	t/h or kW
λ_t	shrinkage σ_t^2/τ_t^2	–
$\bar{\sigma}_t$	prior median of the noise scale	t/h or kW
$\mathbf{m}_t$, $\mathbf{S}_t$	conditional mean and covariance of $\mathbf{d}_t$	t/h or kW
μ_j, ω_j	prior mean and standard deviation of ϑ_j	–
n, N	training surveys (9 in a fold); scored surveys (10)	–
N_{w}, T, D	walkers, sampler steps, sampled dimension	–
$\boldsymbol{\vartheta}$, γ	sampled vector; differential-evolution step	–
$\mathbf{U}$, $\boldsymbol{\Psi}$, $\mathbf{V}$	thin SVD of $\mathbf{Z}$; s_k, $\mathbf{u}_k$ its values, columns	–
$\boldsymbol{\Pi}$	projection $\mathbf{U}\mathbf{U}^{\top}$ on the span of $\mathbf{Z}$	–
$\mathbf{j}$, $\mathcal{I}_t$	sensitivity of $\mathbf{f}_t$ to a scalar; Fisher information	–
$\mathbf{G}$, $\mathbf{b}_t$, $e_{t,h}$	$\mathbf{Z}^{\top}\mathbf{Z}$; engine's plane; its remainder	–
$\mathbf{a}$, ς_c	coefficients of the capacity factor on $\mathbf{Z}$; deviation of the setting	–, mm
$\mathbf{A}_t$, $\boldsymbol{\Lambda}$	linearized Jacobian; prior precision	–
ρ_j	posterior over prior standard deviation	–

The symbols are collected in Table II. The two outputs are measured; the product size, the throw and the speed are not, so an answer to the identification question is an answer for these two outputs only, and a scalar the surveys cannot reach through them is reported as such rather than estimated.

IV. Proposed Method: CrusherBayes

CrusherBayes treats the crusher engine as an exact, cheap simulator and calibrates its fitted scalars jointly with an output-specific structural discrepancy. This section gives the engine, the priors, the discrepancy and its exact marginalization, the posterior predictive, the sampler, the algorithm, its cost and its implementation.

A. The engine as the simulator

For a survey with setting c and feed size f_{80} the engine evaluates a classification and breakage balance on a square-root-of-two sieve grid of 28 classes from 256 mm. The classification function is zero below K_1, one above K_2 and

$$C(d) = 1 - \left(\frac{K_2 - d}{K_2 - K_1}\right)^{K_3} \qquad (K_1 < d < K_2), \tag{1}$$

with, in millimetres and for the calibrated lane of the HP500,

$$\begin{aligned} K_1 &= s_{K_1}\,(0.23\,c + 0.30\,F_{80}),\\ K_2 &= \max\{s_{K_2}\,(12.0 + 0.55\,c + 0.40\,F_{80}),\ 1.05\,K_1\}, \end{aligned} \tag{2}$$

the coefficients being those of Rocha et al. [8] and s_{K_1}, s_{K_2} multipliers whose nominal value is one. Here F_{80} is the 80 % passing size of the feed as the engine represents it, read on its sieve grid, which stops at 256 mm. Rocha et al. write these relations in the measured f_{80}; the engine evaluates them, the t_{10} relation below and the Bond draw at F_{80}. The two agree within 3.1 mm on eight surveys and differ on the two coarsest feeds, where the Rosin–Rammler feed rebuilt from f_{80} has mass above the top of the grid: F_{80} is 118.9 mm for the measured 140 mm of survey 7 and 109.9 mm for the 123 mm of survey 8. The discrepancy design and the comparators use the measured f_{80}. The breakage function follows from the calibrated t_{10} relation

$$t_{10} = \min\{0.44,\ (64 + \Delta t_{10} - 0.12\,c - 0.23\,F_{80})/100\}, \tag{3}$$

through the appearance function of Austin, a two-slope cumulative distribution in the relative size that passes through t_{10} at one tenth of the parent size and is applied column by column to a strictly lower-triangular breakage matrix $\mathbf{B}$. The product mass fractions are

$$\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{B}\mathbf{C})^{-1}\mathbf{f}, \tag{4}$$

with $\mathbf{C}$ the diagonal matrix of the classification of each class and $\mathbf{f}$ the feed, rebuilt from the measured f_{80} as a Rosin–Rammler distribution of uniformity 0.9. Since $\mathbf{B}$ is strictly lower triangular, $\mathbf{I} - \mathbf{B}\mathbf{C}$ has a unit diagonal and the solve is never singular. The throughput depends on the setting and one scalar, through the engine's reduced form of the capacity model of Evertsson [7]: the capacity is Q_{ref} times the discharge opening $c + \mathrm{throw}/2$ over a reference gap of 38 mm, times a hump $\nu\, e^{1-\nu}$ in the ratio ν of the eccentric speed to its optimum. At the fixed throw of 34 mm and speed of 360 rpm, which is the optimal speed of the machine in that form, the hump is one and

$$Q(c) = Q_{\mathrm{ref}}\,\frac{c + 17}{38}. \tag{5}$$

The nominal $Q_{\mathrm{ref}} = 647$ t/h is the number of the base-case capacity Q_T of Rocha et al., but the form is not theirs: their base case is a setting of 38 mm with correction factors, whereas (5) returns 647 t/h at $c = 21$ mm. The Bond draw and the calibrated power are

$$\begin{aligned} P_d &= 10\,W_i\left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}}\right) Q,\\ P_c &= \zeta\,P_d + P_n, \end{aligned} \tag{6}$$

with the sizes in micrometres, W_i the Bond work index of the ore blend, and ζ and P_n the slope and the no-load power

of the power form of Andersen and Napier-Munn [4], at the values of Rocha et al. [8]: ζ fitted by them to the surveys and P_n set to a third of the installed power. Their P_d is the drop-weight energy of the classified feed; the engine's is the Bond draw above, so the pair (ζ, P_n) enters a different net power from the one it was fitted with. The eight scalars $\boldsymbol{\theta} = (Q_{\mathrm{ref}}, W_i, \zeta, P_n, s_{K_1}, s_{K_2}, K_3, \Delta t_{10})$ are the quantities calibrated. Two facts of this structure decide what the surveys can identify, and Section V makes them exact. Throughput depends on Q_{ref} alone, through (5). Power depends on all eight: on the four scalars of the population balance (s_{K_1}, s_{K_2}, K_3, Δt_{10}) only through P_{80} in the Bond term, and on W_i and ζ only through their product. One evaluation of the engine at one operating point takes 1.4 ms (the median of 300 timed evaluations in a single process on the workstation of Section XI), so it is run directly inside the likelihood and no emulator is needed, which removes the emulator error that Higdon et al. [28] must model.

B. Priors

Each scalar has a truncated normal prior around its nominal value, on the logarithmic scale for the positive scalars (Table III). The nominal values are the engine's shipped values: for Q_{ref} the base-case capacity that Rocha et al. [8] give in their Section 2.2, for the other scalars their Table 5 (for K_3 their p. 11; P_n is their third of the installed power, not a fitted value) and, for W_i, the blend of the compact and supercompact itabirite bands (7.2 and 10.6 kWh/t, Table 1 of the same work). The widths are weakly informative choices that bracket the spans quoted there: a logarithmic standard deviation of 0.25 on Q_{ref} (about $\pm 28\,\%$), 0.17 on W_i, 0.23 on ζ, 0.15 on each classification multiplier, a standard deviation of 50 kW on P_n, 0.5 on K_3 and 4 points on Δt_{10}. We found no published range of these fitted scalars for this ore, so the widths are ours, and Section VIII doubles them to measure their effect. The noise scale of each output has $\ln \sigma_t \sim N(\ln \bar{\sigma}_t, 1)$ with medians $\bar{\sigma}_{\mathrm{tph}} = 60$ t/h and $\bar{\sigma}_{\mathrm{kW}} = 15$ kW, about 5 % of the typical level of 1,200 t/h and 300 kW. Writing $\boldsymbol{\vartheta}$ for the sampled vector (the logarithm of each positive scalar, the other scalars as they are, and $\ln \sigma_t$), the log prior is

$$\ln \pi(\boldsymbol{\vartheta}) = -\sum_{j=1}^{8} \frac{(\vartheta_j - \mu_j)^2}{2\omega_j^2} - \sum_{t} \frac{(\ln \sigma_t - \ln \bar{\sigma}_t)^2}{2} + \text{const}, \tag{7}$$

inside the bounds of Table III and minus infinity outside them, with μ_j and ω_j the prior mean and standard deviation of the sampled scale.

C. Discrepancy and exact marginal likelihood

For survey i and output t,

$$y_{t,i} = f_t(\boldsymbol{\theta}; \mathbf{x}_i) + \delta_t(\mathbf{z}_i) + \varepsilon_{t,i}, \quad \varepsilon_{t,i} \sim N(0, \sigma_t^2), \tag{8}$$

$$\delta_t(\mathbf{z}) = d_{t,0} + d_{t,1} z_c + d_{t,2} z_{f_{80}}, \quad d_{t,j} \sim N(0, \tau_t^2), \tag{9}$$

where $\mathbf{z} = (z_c, z_{f_{80}})$ is the pair of setting and feed size standardized with the mean and the population standard deviation of the training surveys, and $\tau_{\mathrm{tph}} = 140$ t/h and $\tau_{\mathrm{kW}} = 36$ kW (about 12 % of the typical level). The discrepancy is the model-form error of Kennedy and O'Hagan [27] reduced to a plane in the two regressors that the surveys vary: a constant that absorbs a level error of the engine and two slopes that absorb a wrong dependence on the setting or the feed size. Given $\boldsymbol{\theta}$ and σ_t the model is linear and Gaussian in $\mathbf{d}_t$, so the discrepancy is integrated out exactly. With $\mathbf{Z}$ the $n \times 3$ matrix of rows $(1, z_{c,i}, z_{f_{80},i})$ of the training surveys and $\mathbf{r}_t = \mathbf{y}_t - \mathbf{f}_t(\boldsymbol{\theta})$ the residual of the engine,

$$\mathbf{y}_t \sim N(\mathbf{f}_t(\boldsymbol{\theta}),\ \boldsymbol{\Sigma}_t), \qquad \boldsymbol{\Sigma}_t = \sigma_t^2 \mathbf{I} + \tau_t^2 \mathbf{Z}\mathbf{Z}^{\top}. \tag{10}$$

The two outputs are treated as independent given $\boldsymbol{\theta}$, so the log likelihood is the sum of the two Gaussian log densities of (10), computed through the Cholesky factor of $\boldsymbol{\Sigma}_t$; the unnormalized log posterior is

$$\ln \pi(\boldsymbol{\vartheta} \mid \mathcal{D}) = \ln \pi(\boldsymbol{\vartheta}) + \text{const} - \sum_{t} \Big[\tfrac{1}{2} \mathbf{r}_t^{\top} \boldsymbol{\Sigma}_t^{-1} \mathbf{r}_t + \tfrac{1}{2} \ln \det \boldsymbol{\Sigma}_t \Big]. \tag{11}$$

The sampler explores the eight scalars and the two noise scales (ten dimensions); the six discrepancy coefficients are never sampled. For each retained sample they follow from the exact Gaussian conditional

$$\mathbf{d}_t \mid \boldsymbol{\theta}, \sigma_t, \mathcal{D} \sim N(\mathbf{m}_t, \mathbf{S}_t), \qquad \mathbf{m}_t = \tau_t^2 \mathbf{Z}^{\top} \boldsymbol{\Sigma}_t^{-1} \mathbf{r}_t, \quad \mathbf{S}_t = \tau_t^2 \mathbf{I} - \tau_t^4 \mathbf{Z}^{\top} \boldsymbol{\Sigma}_t^{-1} \mathbf{Z}, \tag{12}$$

which is how the identifiability table reports the discrepancy coefficients. The low-order form is a choice for nine training surveys: a Gaussian-process discrepancy as in Kennedy and O'Hagan [27] would add a length scale and an amplitude per output to be learned from nine points in two dimensions, whereas the plane has three coefficients per output, a closed-form marginal and a conditional that the identifiability table can report coefficient by coefficient. The price is that a curved model error in the setting is not represented, and it appears as noise in σ_t.

D. Posterior predictive of a held-out survey

For a survey h outside the training set, with $\mathbf{z}_h$ standardized by the training mean and deviation and $\tilde{\mathbf{z}}_h^{\top} = (1, z_{c,h}, z_{f_{80},h})$ its design row, the predictive at one retained sample is Gaussian,

$$y_{t,h} \mid \boldsymbol{\theta}, \sigma_t, \mathcal{D}_{-h} \sim N\big(f_t(\boldsymbol{\theta}; \mathbf{x}_h) + \tilde{\mathbf{z}}_h^{\top} \mathbf{m}_t,\ \sigma_t^2 + \tilde{\mathbf{z}}_h^{\top} \mathbf{S}_t \tilde{\mathbf{z}}_h\big), \tag{13}$$

TABLE III
PRIORS OF THE EIGHT ENGINE SCALARS OF THE HP500 LANE, WITH THE ENGINE'S NOMINAL VALUE (THE PRIOR MEDIAN), THE PRIOR ON THE SAMPLED SCALE AND ITS TRUNCATION RANGE; THE SAMPLED SCALE IS THE NATURAL LOGARITHM FOR THE POSITIVE SCALARS.

Scalar	Unit	Nominal	Prior (sampled scale)	Range
Q_{ref}	t/h	647.00	$\ln x \sim N(6.47,\ 0.25^2)$	300.00 to 1400.0
W_i	kWh/t	9.00	$\ln x \sim N(2.20,\ 0.17^2)$	5.00 to 14.0
ζ	–	1.30	$\ln x \sim N(0.26,\ 0.23^2)$	0.50 to 2.5
P_n	kW	110.00	$N(110.0,\ 50.0^2)$	0.0 to 300.0
s_{K_1}	–	1.00	$\ln x \sim N(0.00,\ 0.15^2)$	0.50 to 2.0
s_{K_2}	–	1.00	$\ln x \sim N(0.00,\ 0.15^2)$	0.50 to 2.0
K_3	–	2.30	$N(2.3,\ 0.5^2)$	1.0 to 4.0
Δt_{10}	points of t_{10}	0.00	$N(0.0,\ 4.0^2)$	−15.0 to 15.0

and the posterior predictive is the mixture of these Gaussians over the retained samples. Its mean is the average of the component means. The central 80 % interval is the 10th to the 90th percentile of 20 draws per retained sample from (13), and the predictive density of the measurement is the average of the component densities, which is the quantity the log score of Section VI uses. The predictive carries three sources of uncertainty: the noise σ_t, the conditional uncertainty of the discrepancy $\mathbf{S}_t$, and the spread of the scalars across the samples. It is a generative model of a survey, which is what distinguishes its band from a residual band built from the spread of held-out errors. The discrepancy at the held-out survey uses the standardization of the training surveys, so a held-out setting outside the training range extrapolates the plane linearly.

E. Sampler and convergence criterion

The sampler is `emcee` 3.1.6 [36] with the differential-evolution move of ter Braak [37] and the snooker move of ter Braak and Vrugt [38], chosen at random in proportions 0.8 and 0.2. The walkers are updated in groups (two for the differential-evolution move, four for the snooker move), each group proposing from the walkers of the others. In the installed implementation, the differential-evolution proposal of a walker $\boldsymbol{\vartheta}$ is

$$\boldsymbol{\vartheta}^* = \boldsymbol{\vartheta} + \gamma\,(\boldsymbol{\vartheta}_a - \boldsymbol{\vartheta}_b), \qquad \gamma = \gamma_0(1 + 10^{-5}\xi),\ \gamma_0 = \frac{2.38}{\sqrt{2D}}, \tag{14}$$

with $\boldsymbol{\vartheta}_a \neq \boldsymbol{\vartheta}_b$ two walkers of the other group, $\xi \sim N(0,1)$ and $D = 10$ the dimension, and the proposal is accepted with probability $\min\{1, \pi(\boldsymbol{\vartheta}^* \mid \mathcal{D})/\pi(\boldsymbol{\vartheta} \mid \mathcal{D})\}$. The snooker move projects the difference of two other walkers on the line through the walker and a third one, with a stretch of 1.7. Both moves use only the log posterior (11), no gradient. The run has 20 walkers (twice the dimension), 7000 steps, the first 1000 discarded and every 10th kept, which retains 12000 samples per posterior, from a fixed seed (42). The walkers start in a ball around the prior centre of 5 % of each prior standard deviation. A posterior counts as converged when

$$T - T_{\mathrm{burn}} \;\geq\; 50\,\hat{\tau}_{\max}, \tag{15}$$

with T the steps, T_{burn} the discarded steps and $\hat{\tau}_{\max}$ the largest over the ten dimensions of the integrated autocorrelation time that `emcee` estimates. A posterior that fails (15) is recorded as not converged and never kept silently. The affine-invariant stretch move of Goodman and Weare [35], the default of `emcee`, was the first choice and fails this criterion on the same posterior (Section VIII).

F. Algorithm, cost and implementation

Algorithm 1 gives one posterior and the predictive of its held-out survey. The cost is dominated by the engine. One log posterior evaluates the engine on the n training surveys (two outputs per evaluation) and factors two $n \times n$ matrices, so a posterior costs about $N_{\mathrm{w}} T n$ engine evaluations, here $20 \times 7000 \times 9 = 1.26 \times 10^6$, and the predictive step adds $(n+1)$ evaluations for each of the 12000 retained samples.

Algorithm 1 CrusherBayes: one posterior and the predictive of the held-out survey h

Require: training surveys $\mathcal{D}_{-h}$, held-out operating point $\mathbf{x}_h$, priors, τ_t, seed
standardize (c, f_{80}) on $\mathcal{D}_{-h}$; build $\mathbf{Z}$ and $\tilde{\mathbf{z}}_h$
draw N_{w} walkers in a ball of 5 % of the prior widths around the prior centre
for $k = 1, \ldots, T$ **do**
  for each group of the walkers **do**
    propose by (14) (probability 0.8) or the snooker move (0.2)
    evaluate the engine on $\mathcal{D}_{-h}$; $\ell \leftarrow$ (11) via Cholesky of $\boldsymbol{\Sigma}_t$
    accept or reject
  end for
end for
estimate $\hat{\tau}_{\max}$; record converged if (15) holds
keep every 10th sample after T_{burn}
for each retained sample **do**
  compute $\mathbf{m}_t, \mathbf{S}_t$ by (12); the engine at $\mathbf{x}_h$
  store mean and variance of (13); draw 20 predictive values
end for
return predictive mean, 10th and 90th percentiles, mixture density at $y_{t,h}$

The linear algebra is $O(n^3)$ per evaluation with $n = 9$ and is negligible beside the engine's 28-class balance. The eleven posteriors of the protocol (ten folds and the fit to all ten surveys) are independent, and the walkers of all of them are evaluated in parallel by a pool of processes; each posterior has its own seed, so its samples do not depend on the number of processes. The implementation is the module `data-pipeline/pipeline/hp500/crusherbayes.py`, on the Python engine `data-pipeline/pipeline/model/crusher/`, which takes the eight scalars through one optional argument whose defaults are the shipped values. The parameters of the method are the prior widths, the discrepancy scales τ_t, the noise prior, the sampler size and the seed, all listed above, and none of them was tuned on the held-out scores. The number of walkers is twice the sampled dimension, and the number of steps was sized to the convergence criterion: on the same posterior the differential-evolution moves give an autocorrelation time of about 50 steps, so 6,000 post-burn steps hold more than the 50 autocorrelation times the criterion asks. The predictive step evaluates the engine once per retained sample at the training surveys and the held-out survey, and the discrepancy conditional (12) needs one more $n \times n$ inverse per sample and output. One run of the protocol yields, for every fold, the predictive summaries of the held-out survey and the convergence diagnostics, and, for the fit to all ten surveys, the posterior summary of every parameter; the scores of the comparators and the terms of the rule for dropping the method are computed from these predictions.

V. Theory

This section states why the method behaves as it does on these surveys: what the data can and cannot tell about each scalar, and what form the predictive mean takes. Propositions 2 to 4 are exact for the model of Section IV, and so is Proposition 1 as a statement about the Fisher information at each value of the scalars; its use as a description of the posterior is a first-order approximation, which the linearized posterior (21) checks against the sampler. Conjecture 1 is stated with the evidence that motivated it, and Remark 1 shows that the results refute it as stated.

A. The information a discrepancy leaves to the scalars

Let $\mathbf{Z} = \mathbf{U}\mathbf{\Psi}\mathbf{V}^\top$ be the thin singular value decomposition of the design, with $\mathbf{U}$ an $n \times 3$ matrix of orthonormal columns and $\mathbf{\Psi} = \operatorname{diag}(s_1, s_2, s_3)$; over the ten surveys the singular values span 2.44 to 3.75. Since $\mathbf{Z}\mathbf{Z}^\top = \mathbf{U}\mathbf{\Psi}^2\mathbf{U}^\top$,

$$\mathbf{\Sigma}_t^{-1} = \frac{1}{\sigma_t^2}\left(\mathbf{I} - \mathbf{U}\mathbf{U}^\top\right) + \mathbf{U}\left(\sigma_t^2\mathbf{I} + \tau_t^2\mathbf{\Psi}^2\right)^{-1}\mathbf{U}^\top, \quad (16)$$

which one checks by multiplying by $\mathbf{\Sigma}_t = \sigma_t^2(\mathbf{I} - \mathbf{U}\mathbf{U}^\top) + \mathbf{U}(\sigma_t^2\mathbf{I} + \tau_t^2\mathbf{\Psi}^2)\mathbf{U}^\top$: the two terms act on orthogonal subspaces.

Proposition 1: Let $\mathbf{j} = \partial\mathbf{f}_t/\partial\vartheta$ be the sensitivity of output t on the training surveys to a sampled scalar ϑ, and $\mathbf{\Pi} = \mathbf{U}\mathbf{U}^\top$ the projection on the column space of $\mathbf{Z}$. The Fisher information of the marginal likelihood (10) about ϑ is

$$\mathcal{I}_t(\vartheta) = \frac{\|(\mathbf{I} - \mathbf{\Pi})\mathbf{j}\|^2}{\sigma_t^2} + \sum_{k=1}^{3}\frac{(\mathbf{u}_k^\top\mathbf{j})^2}{\sigma_t^2 + \tau_t^2 s_k^2}. \quad (17)$$

Proof: For a Gaussian with mean $\mathbf{f}_t(\vartheta)$ and a covariance that does not depend on ϑ, the Fisher information is $\mathbf{j}^\top\mathbf{\Sigma}_t^{-1}\mathbf{j}$. Inserting (16) and writing $\mathbf{j} = (\mathbf{I} - \mathbf{\Pi})\mathbf{j} + \mathbf{U}\mathbf{U}^\top\mathbf{j}$ gives the two terms. The identity holds exactly at every value of ϑ; only reading it as the precision of the posterior is a first-order approximation, exact when $\mathbf{f}_t$ is linear in ϑ. ∎

The proposition says that a discrepancy linear in (c, f_{80}) discounts exactly the part of a sensitivity that is itself an affine function of the setting and the feed size, by the factor $\sigma_t^2/(\sigma_t^2 + \tau_t^2 s_k^2)$, and leaves the orthogonal part untouched. A scalar is identified by the surveys only through the part of its effect that a plane in the two regressors cannot imitate, or through the finite width of the discrepancy prior. With the noise scales at their posterior means on all ten surveys and the values of τ_t, the discount factor for a singular value of 3 is about one tenth on both outputs.

Proposition 1 is the form, for a discrepancy linear in the regressors, of the confounding of calibration parameters and discrepancy that Brynjarsdóttir and O'Hagan [30] and Arendt et al. [31] describe in general: the information about a scalar is cut exactly in the directions the discrepancy can take. The orthogonal prior of Plumlee [33] would forbid the discrepancy the directions of the sensitivities of the model, so that the information in those directions would not be discounted; we keep the prior of Kennedy and O'Hagan, under which Propositions 2 to 4 are exact.

B. Throughput and the capacity base

Proposition 2: On the HP500 lane the vector of engine throughputs of the training surveys lies in the column space of $\mathbf{Z}$, for every value of the scalars: $\mathbf{f}_{\mathrm{tph}}(\boldsymbol{\theta}) = Q_{\mathrm{ref}}\,\mathbf{Z}\mathbf{a}$ with $\mathbf{a} = ((\bar{c}+17)/38,\ \varsigma_c/38,\ 0)^\top$, $\bar{c}$ and ς_c the training mean and deviation of the setting. Consequently the throughput surveys carry no information about the other seven scalars, and their information about $\vartheta = \ln Q_{\mathrm{ref}}$ is

$$\begin{aligned}\mathcal{I}_{\mathrm{tph}}(\ln Q_{\mathrm{ref}}) &= Q_{\mathrm{ref}}^2\,\mathbf{a}^\top\mathbf{Z}^\top\mathbf{Z}\left(\sigma^2\mathbf{I} + \tau^2\mathbf{Z}^\top\mathbf{Z}\right)^{-1}\mathbf{a} \\ &\le \frac{Q_{\mathrm{ref}}^2\|\mathbf{a}\|^2}{\tau^2}, \end{aligned} \quad (18)$$

with σ, τ the throughput scales, so that it vanishes as $\tau \to \infty$.

Proof: By (5), $Q_i = Q_{\mathrm{ref}}(c_i+17)/38$, and $c_i = \bar{c} + \varsigma_c z_{c,i}$, so $Q_i = Q_{\mathrm{ref}}[(\bar{c}+17)/38 + (\varsigma_c/38)z_{c,i}]$, which is row i of $Q_{\mathrm{ref}}\mathbf{Z}\mathbf{a}$. None of the other scalars enters (5). Then $\mathbf{j} = Q_{\mathrm{ref}}\mathbf{Z}\mathbf{a}$ has no component orthogonal to the column space and, by (17) with $\mathbf{Z} = \mathbf{U}\mathbf{\Psi}\mathbf{V}^\top$, $\mathcal{I} = Q_{\mathrm{ref}}^2\mathbf{a}^\top\mathbf{V}\mathbf{\Psi}^2(\sigma^2\mathbf{I} + \tau^2\mathbf{\Psi}^2)^{-1}\mathbf{V}^\top\mathbf{a}$, which is (18). Each factor $s_k^2/(\sigma^2 + \tau^2 s_k^2)$ is below $1/\tau^2$, which gives the bound. ∎

Computed with the engine on the ten surveys, the relative norm of the component of the capacity factors outside the column space of $\mathbf{Z}$ is 2.9×10^{-16}, which is the proposition at machine precision. The capacity base is therefore not identified by the shape of the throughput data, which a plane in the setting reproduces whatever Q_{ref} is; it is identified only because the discrepancy prior penalizes a large plane. The posterior contraction of Q_{ref} must therefore move toward one when τ_{tph} grows and toward smaller values when it shrinks.

C. The predictive mean is a shrunk regression

Proposition 3: Write $\lambda_t = \sigma_t^2/\tau_t^2$, $\mathbf{G} = \mathbf{Z}^\top\mathbf{Z}$, and let $\mathbf{b}_t$ be the least-squares coefficients of the engine output on $\mathbf{Z}$, $\mathbf{f}_t = \mathbf{Z}\mathbf{b}_t + \mathbf{e}_t$ with $\mathbf{Z}^\top\mathbf{e}_t = \mathbf{0}$, and $e_{t,h} = f_t(\boldsymbol{\theta}; \mathbf{x}_h) - \tilde{\mathbf{z}}_h^\top\mathbf{b}_t$ the engine's departure from that plane at the held-out survey. Given the scalars, the predictive mean of (13) is

$$\hat{y}_{t,h} = \tilde{\mathbf{z}}_h^\top\left(\mathbf{G} + \lambda_t\mathbf{I}\right)^{-1}\left(\mathbf{Z}^\top\mathbf{y}_t + \lambda_t\mathbf{b}_t\right) + e_{t,h}. \quad (19)$$

Proof: The push-through identity $\mathbf{Z}^\top(\sigma^2\mathbf{I} + \tau^2\mathbf{Z}\mathbf{Z}^\top)^{-1} = (\sigma^2\mathbf{I} + \tau^2\mathbf{G})^{-1}\mathbf{Z}^\top$ turns (12) into $\mathbf{m}_t = (\mathbf{G} + \lambda_t\mathbf{I})^{-1}\mathbf{Z}^\top(\mathbf{y}_t - \mathbf{f}_t)$. With $\mathbf{Z}^\top\mathbf{f}_t = \mathbf{G}\mathbf{b}_t$, the predictive mean $f_{t,h} + \tilde{\mathbf{z}}_h^\top\mathbf{m}_t = \tilde{\mathbf{z}}_h^\top\mathbf{b}_t + e_{t,h} + \tilde{\mathbf{z}}_h^\top(\mathbf{G} + \lambda_t\mathbf{I})^{-1}(\mathbf{Z}^\top\mathbf{y}_t - \mathbf{G}\mathbf{b}_t)$, and $\mathbf{b}_t - (\mathbf{G} + \lambda_t\mathbf{I})^{-1}\mathbf{G}\mathbf{b}_t = \lambda_t(\mathbf{G} + \lambda_t\mathbf{I})^{-1}\mathbf{b}_t$ gives (19). ∎

Corollary 1: As $\tau_t \to \infty$ ($\lambda_t \to 0$) the predictive mean tends to the ordinary least-squares fit of the survey on the standardized setting and feed size plus the engine's non-affine remainder $e_{t,h}$; for throughput $e_{t,h} = 0$ by Proposition 2, so the limit is the free-form linear fit. As $\lambda_t \to \infty$ it tends to the engine. Between them, the engine enters through a shrinkage target whose weight on each standardized coefficient is $\lambda_t/(g_{kk} + \lambda_t)$ for a diagonal $\mathbf{G}$.

The corollary bounds what CrusherBayes can achieve on these data: given the scalars, its predictive mean is a blend of a linear fit in the two regressors and the engine's best plane, plus the engine's curvature. The values in use are $\lambda_{\mathrm{tph}} = 1.08$ and $\lambda_{\mathrm{kW}} = 1.16$ with the noise at its posterior mean on all ten surveys, so the weight of the engine's plane is about 0.10

on both outputs. On power the engine's best plane over the ten surveys, computed with the scalars at their shipped values (the posterior mean moves them by less than the prior widths, Table IX), has a slope of 8.5 kW per standard deviation of the setting, against -56.0 kW for the measured power, so the shrinkage pulls the slope toward the wrong sign; the method cannot beat a linear fit unless the curvature $e_{t,h}$ carries signal.

D. The work index and the power slope

Proposition 4: The engine power depends on W_i and ζ only through ζW_i, so the likelihood is constant along the curves $\ln\zeta + \ln W_i = \text{const}$, and

$$\frac{\partial \mathbf{f}_{\mathrm{kW}}}{\partial \ln W_i} = \frac{\partial \mathbf{f}_{\mathrm{kW}}}{\partial \ln \zeta} = \zeta\, \mathbf{P}_d. \tag{20}$$

The data inform $\ln\zeta + \ln W_i$ and nothing else about the pair; the marginal of each is the prior along the ridge reweighted by the information on the sum.

Proof: In (6), P_d is linear in W_i, and $P_c = \zeta W_i \cdot 10(P_{80}^{-1/2} - F_{80}^{-1/2})Q + P_n$; P_{80} does not depend on W_i, which on the calibrated lane enters only the Bond term. Differentiating in the logarithms gives the same vector ζP_d. ∎

The finite-difference Jacobians of the engine in $\ln W_i$ and $\ln\zeta$ agree to a relative difference of 9.1×10^{-14}. The no-load power P_n enters power as a constant, $\partial\mathbf{f}_{\mathrm{kW}}/\partial P_n = \mathbf{1}$, which lies in the column space of $\mathbf{Z}$; by Proposition 1 it is discounted like the capacity base and is confounded with the constant of the power discrepancy. The linearized posterior (21) below gives a correlation of -0.27 between W_i and ζ: weaker than the exact ridge suggests, because the two priors have different widths and the information on their sum is small beside the prior precision, so the ridge reweights the prior only slightly.

E. The population balance

Conjecture 1: On these surveys, with throughput and power as the only measured outputs, the four population-balance scalars are not identified: their posterior equals their prior up to the Monte Carlo error of the sampler, and fixing them at their nominal value changes neither the held-out accuracy nor the bands beyond that error.

The evidence that motivated it is threefold. First, by Proposition 2, throughput carries nothing about them. Second, they reach power only through P_{80} in the Bond term, and the Bond draw of the engine varies only from 196 to 237 kW over the ten surveys, against a measured span of 187 to 355 kW. Third, the share of the power sensitivity of each scalar that lies in the column space of $\mathbf{Z}$, and is therefore discounted by Proposition 1, is at least 0.906, the lowest being that of Δt_{10} (Table IV). A second-order statement for all fourteen parameters follows from a Laplace approximation of the linearized model: with $\mathbf{A}_t$ the Jacobian of the mean of output t in the eight sampled scalars and the six discrepancy coefficients, and $\mathbf{\Lambda}$ the diagonal prior precision on the sampled scale,

$$\mathrm{Cov} \approx \Big(\mathbf{\Lambda} + \sum_t \sigma_t^{-2}\mathbf{A}_t^{\top}\mathbf{A}_t\Big)^{-1}, \tag{21}$$

whose diagonal, divided by the prior variance, predicts the square of the ratio of posterior to prior standard deviation on the sampled scale. The sampler reports the ratio (25) on the natural scale, so for a scalar sampled as its logarithm the linearized standard deviation is mapped to the natural scale by the delta method, $\mathrm{sd}_{\mathrm{nat}} = \hat{x}\,\mathrm{sd}_{\log}$ with $\hat{x}$ the posterior mean, and divided by the same natural-scale prior standard deviation that the sampler's ratio uses; Table IV compares the two on that one scale. Evaluated at the posterior mean of the fit to all ten surveys, the linearized posterior gives 0.442 for Q_{ref}, 0.866 for W_i, 0.733 for ζ, 0.891 for P_n and 0.96 to 1.02 for the four population-balance scalars. It matches the sampled ratio within 0.024 on thirteen of the fourteen parameters. The exception is P_n, 0.891 against 0.795: P_n is sampled on its natural scale, so the difference is not one of scale, and the linearization ignores both the truncation of its prior at zero and the curvature of the posterior away from the point of evaluation; the two causes were not separated. A linearized ratio slightly above one has two different causes. For K_3, whose prior is truncated at about -2.6 and $+3.4$ prior standard deviations, the untruncated Gaussian prior of the linearization is compared with the smaller standard deviation of the truncated prior. For s_{K_2} the truncation is too far out to matter; the ratio exceeds one because the delta method maps the logarithmic standard deviation to the natural scale at the posterior mean, which lies above the prior median of one, while the prior's natural-scale deviation is about the median times the logarithmic one; the shift of the mean outweighs the slight contraction of the logarithmic deviation. The ablation that fixes the four scalars (Section VIII) and the three seeds of the full method test the conjecture.

Remark 1 (Conjecture 1 is refuted): Both clauses of Conjecture 1 fail at the level of the Monte Carlo error. The posterior is not the prior: the posterior mean of s_{K_1} is 1.062 to 1.068 over the seeds 42, 43 and 44, and that of s_{K_2} 1.051 to 1.053, against a prior mean of 1.011 for both. These are shifts of 0.33 to 0.37 and 0.26 to 0.27 prior standard deviations, reproduced by every seed (spread of the means 0.007 and 0.002), while the Monte Carlo standard error of a posterior mean is about 0.003 (the posterior standard deviation over the square root of an effective sample size of at least 2256 over the three seeds, 2469 for seed 42). Fixing the four scalars is not neutral either: it lowers the held-out power MAPE by 0.23 points (Section VIII), against a spread of 0.02 points over the three seeds. What the results support is weaker: the four scalars are weakly identified, with a ratio of posterior to prior standard deviation between 0.965 and 1.026 (0.96 to 1.02 linearized), posterior means whose largest shift from the prior mean, that of s_{K_1}, is 0.37 prior standard deviations, and an effect on the held-out scores that is a small fraction of the bootstrap intervals of several points. For identifiability this means that a ratio near one does not mean the surveys are silent: the ratio measures the spread, not the location. Since throughput carries nothing about these scalars (Proposition 2), the shift comes from the power surveys, through a sensitivity that the discrepancy discounts (Proposition 1) but does not remove. The surveys therefore do not calibrate the population balance, but its scalars are not free of them either, and a posterior mean

TABLE IV
THEORY AGAINST THE SAMPLER ON THE FIT TO ALL TEN SURVEYS: THE SHARE OF EACH SCALAR'S POWER SENSITIVITY INSIDE THE COLUMN SPACE OF THE DISCREPANCY DESIGN, THE RATIO OF POSTERIOR TO PRIOR STANDARD DEVIATION PREDICTED BY THE LINEARIZED POSTERIOR (21) AND MAPPED TO THE NATURAL SCALE BY THE DELTA METHOD, AND THE RATIO THE SAMPLER MEASURED ON THE NATURAL SCALE.

Parameter	Share in span (power sensitivity)	Linearized ratio	Sampled ratio	Difference
Q_{ref}	0.999	0.442	0.454	−0.012
W_i	0.999	0.866	0.854	0.012
ζ	0.999	0.733	0.733	0.000
P_n	1.000	0.891	0.795	0.096
s_{K_1}	0.976	0.965	0.965	0.000
s_{K_2}	0.990	1.014	1.026	−0.012
K_3	0.991	1.016	0.992	0.024
Δt_{10}	0.906	0.997	0.990	0.007
$d_{\mathrm{tph},0}$	–	0.859	0.878	−0.019
$d_{\mathrm{tph},1}$ (setting)	–	0.343	0.350	−0.007
$d_{\mathrm{tph},2}$ (f_{80})	–	0.335	0.340	−0.005
$d_{\mathrm{kW},0}$	–	0.926	0.910	0.016
$d_{\mathrm{kW},1}$ (setting)	–	0.401	0.381	0.020
$d_{\mathrm{kW},2}$ (f_{80})	–	0.359	0.362	−0.003

of a classification multiplier is not an estimate of the machine's classification.

F. Conditions

The propositions hold under conditions that the HP500 lane meets and another machine or data set may not. The exactness of Proposition 2 needs a capacity that is affine in the setting, which holds because the throw and the speed are fixed and the speed is at the optimum of the capacity model; with a varying speed the capacity hump of the engine is not affine and the throughput would carry information outside the span. Proposition 3 needs the discrepancy to be linear in the regressors with independent Gaussian coefficients, so a Gaussian-process discrepancy as in Kennedy and O'Hagan [27] would replace the ridge by a kernel smoother with the same structure. Conjecture 1 depends on the outputs measured: a measured product size distribution would give the population-balance scalars a sensitivity that a plane in the setting cannot absorb, and would change it. Identification of the capacity base and the no-load power rests on the finiteness of τ_t, which is the formal content of the statement of Brynjarsdóttir and O'Hagan [30] that the confounding is resolved only by meaningful priors.

VI. EXPERIMENTAL DESIGN

A. Hypotheses stated before the experiments

Four hypotheses were stated before the posteriors were computed: H1 and H2 form the rule for dropping the method, and H3 and H4 are the risks of the method identified when it was proposed.

- H1 Accuracy. The posterior predictive mean beats the setting-only line on power by at least 3.0 points of mean absolute percentage error, or beats the refit capacity base on throughput by the same margin.
- H2 Calibration and sharpness. On both outputs the coverage of the 80 % interval lies within 0.7 to 0.9 and the mean width of the interval is no larger than that of the residual band of the refit engine.
- H3 Identifiability. With ten surveys and two outputs the posterior of most scalars is dominated by the prior.
- H4 Source of the gain. Most of what the method gains over the refit engine comes from the discrepancy, which the setting-only line already approximates.

Algorithm 2 Strict leave-one-survey-out protocol, shared by every method

Require: ten surveys, a method $\mathcal{M}$ that fits on surveys and predicts one
for $h = 1, \ldots, 10$ **do**
  $\mathcal{D}_{-h} \leftarrow$ the nine other surveys
  fit $\mathcal{M}$ on $\mathcal{D}_{-h}$ only (scalars, noise, standardization)
  predict survey h: mean $\hat{y}_h$ and, if $\mathcal{M}$ has one, an 80 % interval
end for
for a residual-band method, $\sigma_h \leftarrow$ sample deviation of the held-out residuals of the nine surveys $\neq h$
score MAPE, RMSE, log density, coverage and width over the ten surveys
bootstrap the ten surveys 20000 times (seed 42) for intervals and paired differences
return the scores with their intervals

The method is dropped as a predictor if H1 fails and H2 fails, both together; whatever the outcome, the identifiability table is reported. With $n = 10$ a difference below about 3.0 points of MAPE is within the noise. The propositions of Section V add consequences that the ablation tests: the capacity-base contraction moves with τ_t and the power accuracy moves toward the linear fit as τ_t grows. Conjecture 1 adds a third, that fixing the population-balance scalars leaves the scores unchanged up to the Monte Carlo error; the results refute it (Remark 1).

B. Data, folds and protocol

The dataset is the ten HP500 surveys of Section III, used as published, with no survey removed. The protocol is strict leave-one-survey-out (Algorithm 2): for each survey a posterior is computed from the other nine alone, and the engine scalars, the noise scales, the discrepancy and the standardization of the regressors are all refit inside the fold, so the held-out survey enters only the prediction. A test of the implementation plants an absurd throughput and power in one survey and checks that the prediction of its own fold does not move and that of another fold, which trains on it, does. One more posterior from all ten surveys gives the identifiability table, making 11 posteriors. Every comparator that fits a quantity on the surveys is refit inside the same folds on the same nine surveys, and every method is scored on the same held-out surveys, so all methods are compared on identical training and test sets. The rule for dropping the method is evaluated by code from the scores alone, and a test of the implementation checks it in every combination of its terms.

C. Methods compared

All comparators are run on the same folds by the same code base as the method; none of them is a reported number of another work. For both outputs: the setting-only line, an ordinary least-squares line of the output on the closed-side setting; the free-form linear fit, an ordinary least-squares fit on the setting and the feed size; the constant mean of the nine training surveys; and the refit engine, the engine at its shipped scalars with the capacity base Q_{ref} of (5) refit by least squares for throughput and the slope and intercept of the power form $P_c = \zeta P_d + P_n$ of (6), with its Bond draw P_d, refit for power, and the same refit plus a ridge residual on the standardized regressors bounded to $\pm 20\,\%$ of the engine value. For power only: Morrell's size-specific energy [15], [16] with the crusher work index M_{ic} refit per fold through the origin, and the same with a refit no-load term, both on the measured f_{80}, the measured throughput and the engine's P_{80}. Two references fit nothing on the surveys: the bare Bond draw at the blend work index, and the engine at its shipped constants. The latter is not the model of Rocha et al. [8] (Section II): it carries their classification, t_{10} and ζ numbers, which they fitted on these ten surveys, and their P_n, a third of the installed power, inside the engine's own capacity form and Bond draw, so on power it is neither a held-out figure nor a reproduction of their fit. On throughput it fits nothing to the surveys: only the base-case capacity of 647 t/h, a simulation in Bruno, and the engine's capacity form, whose reference gap of 38 mm, throw of 34 mm and speed of 360 rpm are machine constants of the code, enter it.

The throughput model of Rocha et al. [8] itself, their Eqs. 10 to 15 with the parameters of their Table 4, is run in two variants. As published, it has $\lambda_{\mathrm{R}} = 0.84$, which the authors fitted on survey 10, so its constants were set with the surveys at hand and it is not a held-out figure; strict, it has λ_{R} refit by least squares on the nine training surveys of each fold (0.862 to 0.906), the one scalar the paper fits. Two of its inputs are not printed by the source: the work index of the blend, for which we use the engine's 9.0 kWh/t between the 7.2 and 10.6 kWh/t of its two ore types, and the moisture of each survey, taken below 1 % so that its correction factor is one. As a check of the implementation, the sum over the ten surveys of the squared relative deviations of the published variant is 0.228, which the source prints as 0.23 (0.216 and 0.249 at the two ore work indices). Their power model, $P_c = \zeta P_d + P_n$ with the drop-weight power $P_d = Q \sum_i E_{\mathrm{cs},i} C_i w_i$ of their Eq. 8, cannot be run from the source, because two of its inputs are not printed: the mass fraction w_i of each size class in the crusher, since the source tabulates only the f_{80} of each survey and gives the size distributions as figures, and the specific energy E_{cs} at the t_{10} of each survey, since their energy table stops at $t_{10} = 30\,\%$ while their Eq. 5 with their Table 5 gives 27.2 to 47.4 % on the ten surveys, above the table on 9 of them, and the source gives no rule to extend it. We do not guess either input, so their power model is not among the comparators.

The residual band of a comparator is its prediction plus or minus $1.2816\,\sigma_h$, with σ_h the sample standard deviation of the held-out residuals of the other nine surveys, so that a survey's own residual never enters the width of its band.

D. Metrics

For output t, held-out predictions $\hat{y}_{t,i}$ and measurements $y_{t,i}$ over the $N = 10$ surveys,

$$\mathrm{MAPE}_t = \frac{100}{N} \sum_{i=1}^{N} \frac{|\hat{y}_{t,i} - y_{t,i}|}{y_{t,i}},$$
$$\mathrm{RMSE}_t = \Big(\frac{1}{N} \sum_{i=1}^{N} (\hat{y}_{t,i} - y_{t,i})^2 \Big)^{1/2}. \tag{22}$$

The log predictive density is the mean over surveys of the log density of the measurement under the held-out predictive,

$$\mathrm{LPD}_t = \frac{1}{N} \sum_{i=1}^{N} \ln \Big[\frac{1}{M} \sum_{m=1}^{M} \phi\big(y_{t,i}; \mu_i^{(m)}, v_i^{(m)}\big) \Big], \tag{23}$$

with ϕ the Gaussian density and $(\mu_i^{(m)}, v_i^{(m)})$ the mean and variance of (13) at retained sample m; for a comparator it is the Gaussian density with its residual-band deviation. It is a strictly proper score [40], in log units per t/h or per kW, higher being better. The coverage and the mean width of the 80 % interval $[\ell_{t,i}, u_{t,i}]$ are

$$\kappa_t = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{\ell_{t,i} \le y_{t,i} \le u_{t,i}\},$$
$$w_t = \frac{1}{N} \sum_{i=1}^{N} (u_{t,i} - \ell_{t,i}), \tag{24}$$

read together as the calibration and the sharpness of the interval [39]. The identifiability of a parameter is the ratio

$$\rho_j = \mathrm{sd}_{\mathrm{post}}(\theta_j) / \mathrm{sd}_{\mathrm{prior}}(\theta_j) \tag{25}$$

on the natural scale, read as identified up to 0.5, partly identified up to 0.9 and not contracted above 0.9; a ratio near one says that the posterior is as wide as the prior, not that it sits where the prior does. MAPE and RMSE measure the point prediction, the log density measures the whole predictive distribution, and coverage with width measure the interval; a method can be good on one and poor on another, which is why the hypotheses H1 and H2 are stated on different metrics. The prior standard deviation of a positive scalar is computed on the natural scale by sampling its truncated log-normal prior.

E. Statistics

The ten held-out surveys are the unit of resampling. The 95 % interval of a MAPE or an RMSE is the percentile interval of 20000 bootstrap resamples of the ten surveys [42], with a fixed seed (42); a paired difference of MAPE between CrusherBayes and a comparator is bootstrapped with the same resamples, so that the correlation of the two errors across surveys is kept, and reported with the fraction of resamples in which CrusherBayes has the lower MAPE and the number of surveys on which it has the lower error. A percentile interval of a mean over ten surveys tends to cover less than its nominal

level, so the intervals are, if anything, too narrow. The interval of a coverage k/N is the Wilson score interval [43],

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \qquad z = 1.96. \tag{26}$$

The posterior uncertainty of each parameter is its posterior standard deviation. The Monte Carlo error of the sampler is measured by repeating the whole protocol with the seeds 43 and 44. With ten surveys these intervals are wide, and they are reported so that a difference is read against them rather than as a ranking of point values. No significance test is used to accept or reject a hypothesis.

F. Ablations planned

Each variant changes one component and reruns the whole protocol (eleven posteriors) with the same code, the same folds, sampler size and seed. The discrepancy: its scale reduced to 10^{-6} of τ_t, so the likelihood is the engine plus noise. The engine scalars: all eight held at their nominal value (prior standard deviation 10^{-4} of the method's, on the sampled scale), so that only the discrepancy and the noise are learned. The shared population-balance scalars: s_{K_1}, s_{K_2}, K_3 and Δt_{10} held at their nominal value while capacity, work index and power form are sampled. The priors: every engine prior width doubled. The discrepancy scale: τ_t halved and doubled. The sampler: the stretch move in place of the differential-evolution moves. The seeds 43 and 44 of the full method measure the Monte Carlo error against which every variant is read. Each variant is scored by the same code as the method, so that the same scores, the same contraction table and the same rule for dropping the method are computed for it.

VII. Results

A. Held-out accuracy against the published methods

Table V gives every method on the same folds with its bootstrap interval, and Fig. 1 draws the intervals. On throughput the posterior predictive mean has a MAPE of 12.00 % (95 % interval 7.7 to 15.9), against 12.09 % for the refit capacity base, 12.18 % for the free-form fit, 13.45 % with the bounded residual, 13.67 % for the setting-only line and 18.29 % for the constant mean. The engine at its shipped constants reaches 11.55 %; on throughput only its capacity form and the base-case capacity $Q_{\mathrm{ref}} = 647$ t/h of a Bruno simulation enter, and nothing of it is fitted to the surveys, so this row is a held-out figure, with no band. The published throughput model of Rocha et al. reaches 10.62 % (4.5 to 17.7), the lowest point value on throughput, but its λ_{R} was fitted on survey 10 and its work index is not printed by the source: at the work indices of the two ore types it gives 11.86 % and 10.44 %. With λ_{R} refit in each fold it reaches 14.65 % (10.6 to 19.2), the held-out figure of its form. The label-shuffle control, the refit engine with its bounded residual fitted to one seeded permutation of the measurements, gives 25.04 % on throughput and 40.19 % on power, above every method of Table V that fits the surveys, so a residual fitted to the wrong surveys does not help and the gains read below come from the measurements. The intervals of the throughput methods overlap almost entirely. On power the posterior predictive mean has a MAPE of 14.31 % (9.0 to 19.5), against 12.78 % (7.4 to 18.4) for the setting-only line, 14.03 % for the free-form fit and 15.21 % for Morrell with a no-load term. The engine-based power models are far behind: 25.94 % for the refit power form, 25.37 % with the bounded residual, 26.88 % for the Bond draw, 38.08 % for Morrell through the origin and 47.58 % for the engine at its shipped constants, against 24.48 % for the constant mean.

Figure 1 sorts the methods by their held-out MAPE. On power it shows two groups: four methods whose intervals overlap one another almost entirely (the setting-only line, the free-form fit, CrusherBayes and Morrell with a no-load term), and the engine-based forms and the constant mean, whose intervals lie wholly above the point value of CrusherBayes. On throughput the intervals of all nine methods overlap; the constant mean has the highest point value and the widest interval, and the published model of Rocha et al. the lowest point value with the second widest interval. The figure does not show the paired structure: two methods with overlapping intervals can still differ consistently survey by survey, which Table VI tests. Nor does it show which rows had constants set with the surveys at hand, which the caption of Table V gives. What follows from the figure alone is that, with ten surveys, the marginal intervals resolve no ranking within the first group on power or among the methods on throughput, while they separate the two groups on power; the paired differences resolve one more pair on throughput, read below.

The paired comparison (Table VI) settles H1. Against the setting-only line on power the difference of MAPE is 1.54 points (95 % interval −1.11 to 4.10), CrusherBayes has the lower error on 5 of the ten surveys, and the gain the hypothesis asked for, 3.0 points, lies outside the interval. Against the refit capacity base on throughput the difference is −0.09 points (−6.30 to 6.06), a tie. H1 does not hold. The comparison does show where the method gains: against every engine-based power model except Morrell with a no-load term the difference is large and its interval excludes zero, −11.63 points against the refit power form (−18.25 to −4.47) and −12.56 points against the Bond draw (−21.98 to −3.58), and against the constant mean on power −10.16 points (−19.32 to −0.72). Morrell with a no-load term, which also uses the engine's P_{80}, ties CrusherBayes at −0.90 points (−8.71 to 6.89). The gain is over the engine's own power forms, not over the simple lines. On throughput one pair is separated: against the capacity base with a bounded residual the difference is −1.45 points (−2.06 to −0.76), with the lower error on 9 of the ten surveys. Both methods correct the engine with a plane in the setting and the feed size, CrusherBayes through the shrunk regression of Proposition 3 and the bounded residual through a ridge fit clipped at a fifth of the engine's value, so the paired difference isolates the way the plane is fitted; it is a small, consistent gain over a correction that the ten surveys show to be no better than the refit engine itself. Against the published throughput model of Rocha et al. the difference is 1.38 points (−5.81 to 7.93), CrusherBayes having the lower error on 4 of the ten surveys, and against the same model with λ_{R} refit in each fold it is −2.65 points (−7.63 to 1.98), on 6 surveys: both are ties. The

TABLE V

HELD-OUT SCORES OF CRUSHERBAYES AND OF EVERY COMPARED METHOD ON THE TEN HP500 SURVEYS (STRICT LEAVE-ONE-SURVEY-OUT, THE SAME FOLDS FOR ALL), WITH 95 % PERCENTILE BOOTSTRAP INTERVALS OF THE MAPE OVER THE SURVEYS, AND THE COVERAGE OF THE 80 % INTERVAL WITH ITS WILSON INTERVAL AND THE MEAN WIDTH FOR THE METHODS THAT GIVE A BAND. TWO ROWS FIT NOTHING IN THE FOLDS AND HAD CONSTANTS SET WITH THESE SURVEYS AT HAND: THE ENGINE AT ITS SHIPPED CONSTANTS ON POWER, WHOSE CLASSIFICATION, t_{10} AND ζ ROCHA ET AL. FITTED ON THEM, AND THE PUBLISHED THROUGHPUT MODEL OF ROCHA ET AL., WHOSE λ THEY FITTED ON SURVEY 10; THE ENGINE AT ITS SHIPPED CONSTANTS ON THROUGHPUT FITS NOTHING TO THE SURVEYS.

Target	Method	Fitted on the training surveys	MAPE (%)	95% interval	RMSE	Coverage [95% interval]	Width
Power (kW)							
	CrusherBayes	eight scalars, noise, discrepancy	14.31	[9.0, 19.5]	44.7	0.9 [0.60, 0.98]	117.7
	Setting-only line	intercept, slope	12.78	[7.4, 18.4]	44.6	0.8 [0.49, 0.94]	119.6
	Free-form linear fit	intercept, two slopes	14.03	[8.4, 19.7]	48.7	0.8 [0.49, 0.94]	129.8
	Morrell with no-load term	M_{ic}, no-load power	15.21	[6.4, 24.9]	46.5	–	–
	Refit power form	ζ, P_n	25.94	[17.3, 34.9]	78.5	0.9 [0.60, 0.98]	209.8
	Refit power form with bounded residual	ζ, P_n, ridge residual	25.37	[16.0, 37.9]	92.1	–	–
	Morrell through the origin	M_{ic}	38.08	[24.4, 54.1]	111.7	–	–
	Constant mean	the mean	24.48	[15.2, 35.2]	68.5	–	–
	Bond draw, blend work index	none	26.88	[20.0, 33.2]	93.2	–	–
	Engine at its shipped constants	none	47.58	[24.6, 73.5]	132.8	–	–
Throughput (t/h)							
	CrusherBayes	eight scalars, noise, discrepancy	12.00	[7.7, 15.9]	168.6	0.8 [0.49, 0.94]	439.1
	Refit capacity base	Q_{ref}	12.09	[6.0, 18.7]	176.7	0.8 [0.49, 0.94]	473.8
	Free-form linear fit	intercept, two slopes	12.18	[7.6, 16.3]	174.4	0.8 [0.49, 0.94]	468.5
	Capacity base with bounded residual	Q_{ref}, ridge residual	13.45	[8.6, 17.7]	189.5	–	–
	Setting-only line	intercept, slope	13.67	[9.7, 18.0]	177.8	0.8 [0.49, 0.94]	478.8
	Constant mean	the mean	18.29	[9.2, 28.5]	258.5	–	–
	Engine at its shipped constants	none	11.55	[7.4, 16.2]	171.4	–	–
	Rocha et al., as published	none (λ fitted by the authors on survey 10)	10.62	[4.5, 17.7]	186.3	–	–
	Rocha et al., λ refit	λ	14.65	[10.6, 19.2]	190.4	–	–

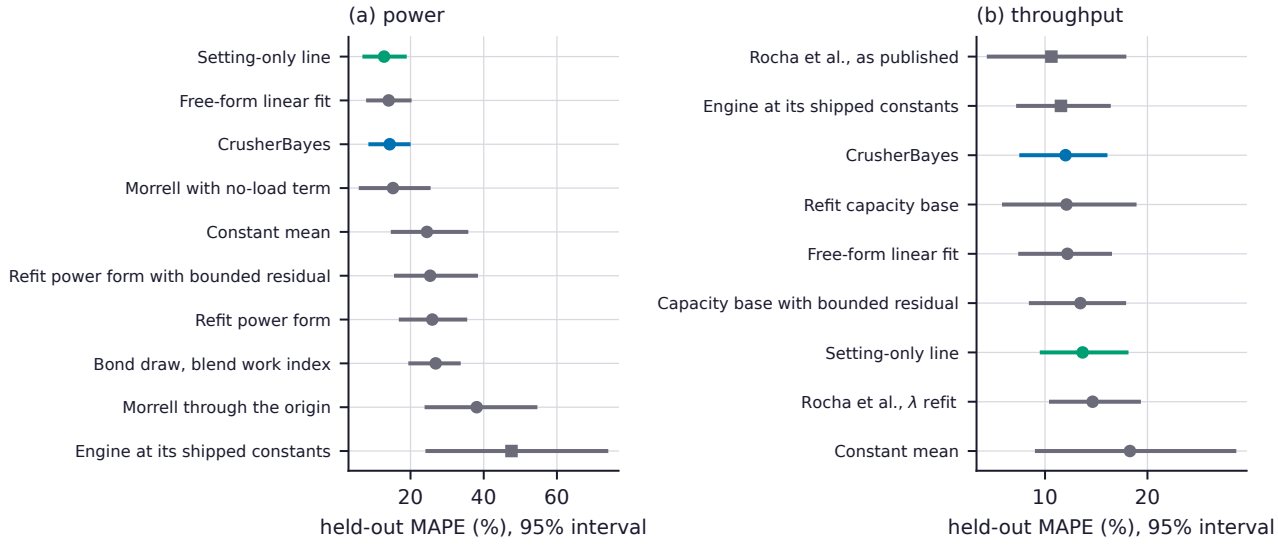


Fig. 1. Held-out mean absolute percentage error of every method on the ten HP500 surveys with its 95 % bootstrap interval, for (a) power and (b) throughput; CrusherBayes in blue, the setting-only line in green, the engine at its shipped constants and the published model of Rocha et al. (no fit in the folds) as squares.

published variant is not held out on survey 10, on which its one fitted number was set, and its point value depends on a work index the source does not print, so the tie is the reading the data support. The closest published model of this crusher is therefore no better and no worse than CrusherBayes on throughput, and its power model could not be run (Section VI).

B. Calibration and sharpness

The 80 % interval of CrusherBayes covers 0.8 of the surveys on throughput (Wilson interval 0.49 to 0.94) and 0.9 on power (0.60 to 0.98), both inside 0.7 to 0.9. Its mean width is 439.12 t/h on throughput, 7.3 % narrower than the residual band of the refit engine (473.76 t/h) at the same coverage, and 117.74 kW on power, 43.9 % narrower than that band (209.79 kW) at the same coverage of 0.9. H2 holds. Against the simple models the picture is different: on power the setting-only line has a mean width of 119.60 kW at a coverage of 0.8, and the free-form fit 129.82 kW at 0.8, so the band of CrusherBayes has about the width of the line's band with a coverage higher by one survey. On throughput all four banded methods cover 0.8 and CrusherBayes has the narrowest band. The Wilson intervals of the coverages, 0.60 to 0.98 and 0.49 to 0.94, overlap almost entirely: with ten surveys the coverages do not distinguish a calibrated band from a slightly

TABLE VI
PAIRED DIFFERENCES OF HELD-OUT MAPE, CRUSHERBAYES MINUS EACH COMPARATOR (NEGATIVE FAVOURS CRUSHERBAYES), WITH THE 95 % PAIRED-BOOTSTRAP INTERVAL, THE FRACTION OF RESAMPLES FAVOURING CRUSHERBAYES AND THE SURVEYS ON WHICH IT HAS THE LOWER ERROR.

Target	Comparator	Δ MAPE (points)	95% interval	$P(\Delta < 0)$	Surveys won
Power					
	Setting-only line	1.54	[−1.11, 4.10]	0.130	5 of 10
	Free-form linear fit	0.28	[−3.36, 3.77]	0.432	6 of 10
	Morrell with no-load term	−0.90	[−8.71, 6.89]	0.584	5 of 10
	Refit power form	−11.63	[−18.25, −4.47]	0.999	7 of 10
	Refit power form with bounded residual	−11.06	[−21.30, −3.63]	1.000	8 of 10
	Morrell through the origin	−23.77	[−39.44, −9.70]	1.000	9 of 10
	Constant mean	−10.16	[−19.32, −0.72]	0.983	6 of 10
	Bond draw, blend work index	−12.56	[−21.98, −3.58]	0.998	7 of 10
	Engine at its shipped constants	−33.27	[−56.55, −12.16]	1.000	8 of 10
Throughput					
	Refit capacity base	−0.09	[−6.30, 6.06]	0.503	5 of 10
	Free-form linear fit	−0.18	[−1.36, 1.01]	0.623	6 of 10
	Capacity base with bounded residual	−1.45	[−2.06, −0.76]	1.000	9 of 10
	Setting-only line	−1.67	[−4.61, 2.23]	0.825	8 of 10
	Constant mean	−6.29	[−15.85, 2.19]	0.913	5 of 10
	Engine at its shipped constants	0.45	[−3.44, 4.49]	0.416	6 of 10
	Rocha et al., as published	1.38	[−5.81, 7.93]	0.341	4 of 10
	Rocha et al., λ refit	−2.65	[−7.63, 1.98]	0.859	6 of 10

TABLE VII
HELD-OUT SCORES OF THE METHODS THAT GIVE AN 80 % BAND, OVER THE TEN SURVEYS: MAPE, RMSE, MEAN LOG PREDICTIVE DENSITY, COVERAGE AND MEAN WIDTH OF THE INTERVAL, FOR EACH OUTPUT.

Target	Method	MAPE (%)	RMSE	Mean log density	Coverage	Width
Throughput (t/h)						
	CrusherBayes	12.00	168.57	−6.678	0.8	439.1
	Setting-only line	13.67	177.75	−6.709	0.8	478.8
	Refit engine with residual band	12.09	176.67	−6.896	0.8	473.8
	Free-form linear fit	12.18	174.42	−6.647	0.8	468.5
Power (kW)						
	CrusherBayes	14.31	44.68	−5.346	0.9	117.7
	Setting-only line	12.78	44.56	−5.341	0.8	119.6
	Refit engine with residual band	25.94	78.52	−5.847	0.9	209.8
	Free-form linear fit	14.03	48.68	−5.399	0.8	129.8

miscalibrated one. The log predictive density (Table VII) agrees, and it does not rank CrusherBayes first: on power the line has −5.34, CrusherBayes −5.35, the free-form fit −5.40 and the refit engine −5.85; on throughput the free-form fit has −6.65, CrusherBayes −6.68, the line −6.71 and the refit engine −6.90. On each output the first three are close to one another and the refit engine, with its residual band, is last.

Figure 2 draws the bands survey by survey. On power the band of CrusherBayes misses survey 4 and the residual band of the refit engine misses survey 7; on throughput CrusherBayes misses surveys 1 and 3 and the refit engine surveys 1 and 6. The two methods therefore fail on different surveys, so the equal coverages do not mean the same surveys are covered. The figure also shows where the width comes from: on power the half-widths of CrusherBayes run from 38 to 69 kW and those of the residual band from 89 to 111 kW, so its band is narrower on every survey, while on throughput the two ranges, 165 to 277 and 158 to 253 t/h, overlap. The figure does not show whether a miss is a model error or a measurement error of that survey; for power, whose values the source estimated from electric current, the two cannot be told apart. What follows is that the gain in sharpness is a power result, and that on throughput the bands of all methods are set by the same scatter of the surveys around a line in the setting. Figure 3 draws the same predictions against the setting.

Figure 3 makes the structure of Section V visible. The engine at its shipped constants puts the power at 365 to 419 kW, rising with the setting, while the measured power falls from 355 to 187 kW as the setting widens; the held-out mean of CrusherBayes follows the measurements, because the discrepancy has taken over the setting dependence. On throughput the engine, the line and CrusherBayes all follow the rise of the throughput with the setting, and the scatter around it, which neither the setting nor the feed size explains, sets the width of every band. The figure does not show the feed size, the second regressor of the discrepancy, so two surveys at one setting can differ in their prediction for a reason the horizontal axis hides, and it does not show the comparators other than the line and the engine. What follows is that the setting alone carries most of the dependence of both outputs that the methods capture, which is why a line in the setting is so hard to beat.

TABLE VIII
THE TERMS OF THE RULE FOR DROPPING THE METHOD AS A PREDICTOR, EVALUATED ON THE HELD-OUT SCORES OF THE TEN SURVEYS; THE REFERENCE IS THE SETTING-ONLY LINE FOR POWER AND THE REFIT CAPACITY BASE FOR THROUGHPUT, AND THE GAIN IS COMPUTED FROM THE UNROUNDED ERRORS, AS IN TABLE VI.

Term	Throughput	Power
MAPE of CrusherBayes (%)	12.00	14.31
MAPE of the reference (%)	12.09	12.78
Gain over the reference (points)	0.09	−1.54
Gain of at least 3 points	no	no
Coverage	0.8	0.9
Coverage outside 0.7 to 0.9	no	no
Mean width (t/h, kW)	439.1	117.7
Mean width of the residual band	473.8	209.8
Band wider than the residual band	no	no

C. The dropping rule

Table VIII applies the rule stated before the run. Condition (a) holds: the gain in power MAPE over the setting-only line is −1.54 points and the gain in throughput MAPE over the refit capacity base is 0.09 points, both below 3.0. Condition (b) fails: the coverages 0.8 and 0.9 are inside the interval 0.7 to 0.9, and the bands are narrower than the residual band on both outputs. Both conditions are needed to drop the method, so the rule does not drop it. The rule therefore retains a method whose band meets the coverage criterion and is narrower than the residual band, and that does not improve the point prediction. Every variant of the ablation gives the same outcome: condition (a) holds and condition (b) fails in all of them, including the variant without the discrepancy, whose coverage of 0.7 is at the lower edge of the interval and whose band remains narrower than the residual band.

D. What the surveys identify

Table IX compares the prior with the posterior of every parameter, fitted to all ten surveys, and Figs. 4 and 5 draw it. The table gives both the nominal value, which is the prior median, and the prior mean, which differs from it for the scalars sampled as logarithms and for the truncated priors. Of the 16 parameters, 7 are identified, 4 are partly identified and 5 are not contracted. The identified ones are the capacity base Q_{ref} (ratio 0.454, a contraction that comes from the finite scale of the discrepancy prior and not from the shape of

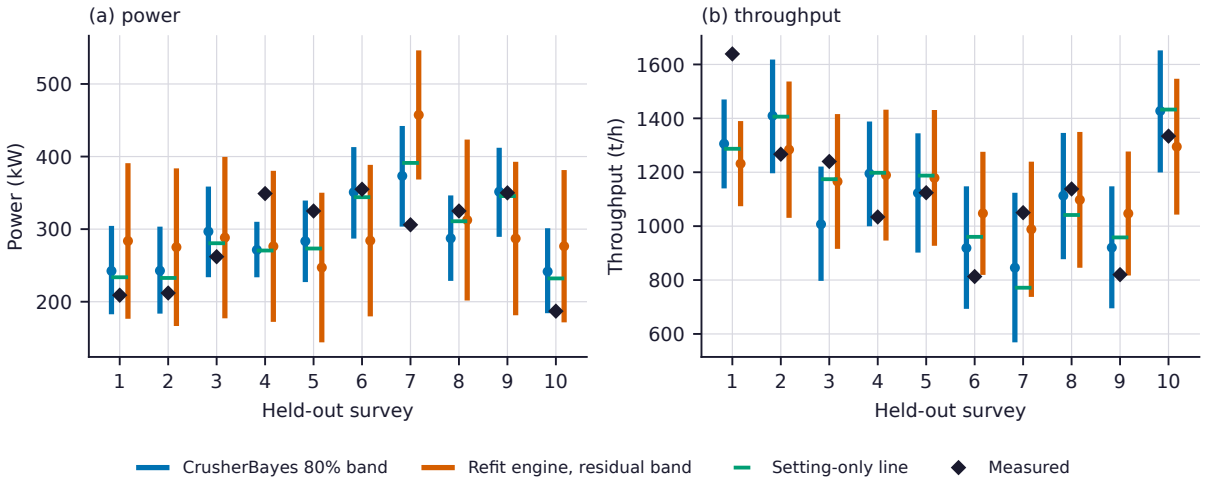


Fig. 2. Held-out 80 % predictive interval against the measured value, per survey, for (a) power and (b) throughput: the posterior predictive of CrusherBayes in blue, the residual band of the refit engine in orange, the setting-only line as a green dash and the measurement as a black diamond; every interval is computed without the survey it is drawn for.

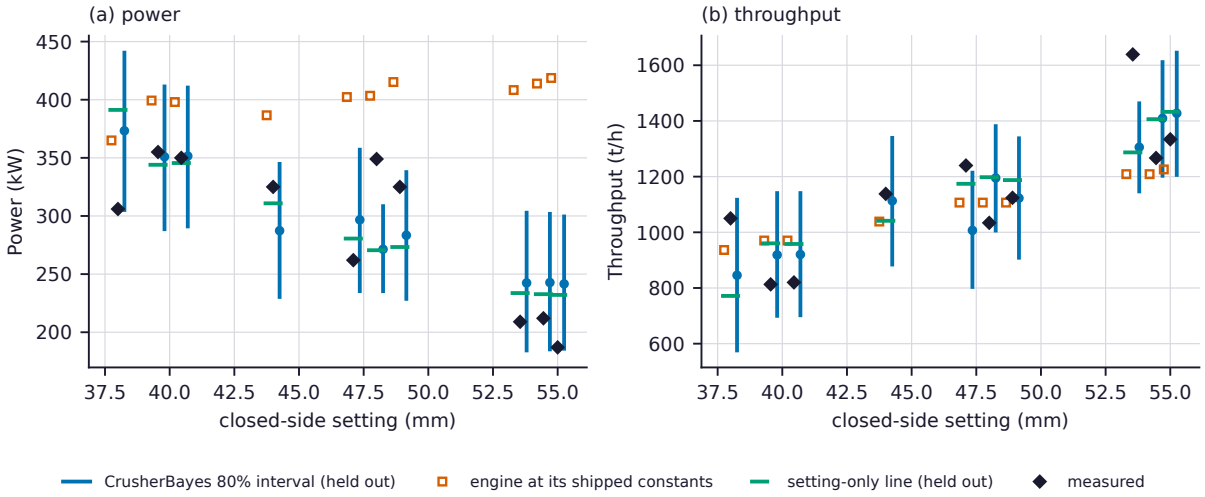


Fig. 3. The two outputs of the ten surveys against the closed-side setting: the measurement, the held-out interval and mean of CrusherBayes, the held-out setting-only line and the engine at its shipped constants, for (a) power and (b) throughput; surveys at one setting are spread by 0.9 mm.

the throughput data, by Proposition 2), the two noise scales (0.301 and 0.332) and the four discrepancy slopes (0.340 to 0.381). Partly identified are ζ (0.733), P_n (0.795), W_i (0.854) and the constant of the throughput discrepancy (0.878). Four of the eight engine scalars are not contracted: s_{K_1} (0.965), s_{K_2} (1.026), K_3 (0.992) and Δt_{10} (0.990), and neither is the constant of the power discrepancy (0.910). Not contracted is not the same as unmoved: the posterior means of s_{K_1} and s_{K_2}, 1.07 and 1.05, lie 0.37 and 0.26 prior standard deviations above their prior mean of 1.011 (Remark 1). The posterior of Q_{ref} is 646.35 $\pm$ 75.74 t/h against a nominal 647 t/h: the surveys narrow the capacity base and leave its median where it was. H3 holds, in the form that the theory predicts: the scalars that only power can inform, through a sensitivity that lies almost wholly in the span of the discrepancy, keep the width of their prior, and the linearized posterior of Table IV gives the same reading (identified, partly identified or not contracted) as the sampler for every parameter.

Figure 4 draws each posterior as a normal density over its prior. It shows three patterns. The capacity base, the noise scales and the four discrepancy slopes are visibly narrower than their priors. The power-form scalars ζ, W_i and P_n are somewhat narrower and shifted below their nominal values, along the ridge of Proposition 4. The four population-balance scalars have the width of their prior, and the two classification multipliers are shifted to the right of the prior mean (dashed line), the shift Remark 1 reads. The figure does not show the shape of the posterior beyond its first two moments, nor the correlation along the ridge. Figure 5 orders the same ratios: the discrepancy slopes and the noise scales come first, then the capacity base, the power-form scalars and the constant of the throughput discrepancy, and last the population-balance scalars and the constant of the power discrepancy, all near one. It does not show the location shifts. What follows from the

two figures is that the surveys inform what the discrepancy cannot imitate (the noise, the slopes of the plane, and the capacity base through the finite τ_{tph}) and leave the scalars of the population balance with the width of their prior.

E. The discrepancy and the shrinkage

The discrepancy slope on the setting is large for both outputs. For power it is -57.07 ± 13.70 kW per standard deviation of the setting (about -9.6 kW/mm, with 5.9 mm the standard deviation of the setting over the ten surveys), and for throughput 100.84 ± 49.00 t/h per standard deviation (about 17.0 t/h per mm). The measured power falls as the setting widens (correlation -0.82), whereas the engine power at its shipped constants rises with it (correlation 0.79, from 365 to 419 kW); the negative slope of the power discrepancy compensates the setting dependence of the engine. Figure 6 evaluates the closed form (19) with the scalars at their nominal value over the shrinkage λ. On throughput the held-out MAPE is 12.18 % at $\lambda = 0$, which is the free-form fit as Corollary 1 states, and 11.53 % for a large λ, which is the engine at its shipped constants; between them it dips to 10.78 % near $\lambda = 17.8$, and at the λ of the method it is 11.91 %, against 12.00 % for the sampled method. A stronger shrinkage would have fit these folds better, but choosing λ on the held-out scores would tune the method on its test set, and the dip of about one point is inside the bootstrap interval of every throughput method. On power it rises from 13.12 % at $\lambda = 0$ to 47.17 % for a large λ, with 15.39 % at the λ of the method against 14.31 % sampled: shrinking toward the engine costs accuracy on power, as its wrong-signed slope predicts. The figure evaluates the closed form at the nominal scalars only, so it does not show the effect of sampling them, nor the band or its coverage; what follows from it is that the accuracy of the method is set by the shrinkage of a plane, which is the content of Proposition 3, and not by the population balance.

F. Convergence

All 11 posteriors meet the criterion (15) (Table X and Fig. 7): the largest autocorrelation time is between 48.0 and 84.8 steps and the 6000 post-burn steps are 70.7 to 125.0 times it, with an acceptance fraction between 0.171 and 0.204. The fold without survey 4 has the slowest chain. Across the three seeds the throughput MAPE varies by 0.01 points and the power MAPE by 0.02 points, and the power band width by 0.1 kW, so the Monte Carlo error of the sampler is below every difference between methods that the text reads as a gain or a loss, and below the smallest difference it reports, the tie of -0.09 points on throughput against the refit capacity base. The retained samples of a fold, 12000, are far more than the summaries need: the posterior means and standard deviations of the identifiability table and the percentiles of the predictive interval are averages over at least 70.7 effective draws per walker. The ratios of posterior to prior standard deviation of the three seeds differ by at most 0.039, on s_{K_1}, and by at most 0.023 on every other parameter; no parameter changes its reading in Table IX from one seed to another. Figure 7 shows the slowest direction of each chain only: it does not show how fast each parameter mixes on its own, nor the seeds 43 and 44, whose largest autocorrelation times are in Table XI. What follows from it is that the differential-evolution moves pass the criterion in every posterior and the stretch move in none.

VIII. Ablation

Table XI and Fig. 8 give the ablation; every row is a full rerun of the eleven posteriors with the same code. The seed rows bound the Monte Carlo error: 11.99 and 11.99 % on throughput and 14.29 and 14.31 % on power, against 12.00 and 14.31 % for seed 42.

The discrepancy carries the power prediction. Without it, the power MAPE rises to 27.15 %, back to the level of the refit engine (25.94 %), the band widens to 190.21 kW and the coverage falls to 0.7 on both outputs; the log density of power drops from -5.35 to -5.82. Throughput hardly changes (11.98 %), because the engine throughput is already a line in the setting (Proposition 2); the capacity base contracts more without the discrepancy (ratio 0.195 against 0.454), as Proposition 2 predicts when $\tau_{\mathrm{tph}} \to 0$. H4 holds.

The engine scalars contribute little. With all eight held at their nominal value the throughput MAPE is 11.95 % and the power MAPE 15.99 % against 14.31 %, with a power band of 124.50 kW at a coverage of 0.7; calibrating the scalars improves power by 1.68 points and does not change throughput. The discrepancy slopes are the same with and without the scalars (-55.8 against -57.1 kW on power).

The population-balance scalars contribute little, but not nothing. Held at their nominal value, they leave the throughput MAPE at 11.99 %, the power MAPE at 14.08 % and the power band at 117.70 kW, all close to the full method, but the power coverage falls from 0.9 to 0.8, one survey of ten, which every seed of the full method keeps at 0.9 (Table XI); the bands of Conjecture 1 are therefore not unchanged either, though the change is the smallest step ten surveys can show. The contraction of the other scalars hardly moves (Q_{ref} 0.444, ζ 0.687). The power MAPE is 0.23 points lower than with the four scalars sampled, a difference about 11 times the seed spread of 0.02 points, so it is a real effect of sampling them and not Monte Carlo error. It refutes the score clause of Conjecture 1, as the location shift of s_{K_1} and s_{K_2} refutes its identification clause (Remark 1). The effect is small beside the survey-level uncertainty: the bootstrap interval of the power MAPE of the method spans 9.0 to 19.5 %.

The priors matter for identification and not for accuracy. Doubling every engine prior width gives 12.08 % and 14.43 %, with a power band of 119.24 kW; the ratios are taken against the wider prior, so Q_{ref} falls to 0.314 while W_i (0.889) and ζ (0.776) remain partly identified, and one fold of eleven (without survey 4) fails the convergence criterion, with an autocorrelation time of 141.7 steps: the wider prior opens the ridge of Proposition 4.

The discrepancy scale moves the method along the shrinkage of Proposition 3. Halving τ_t shrinks harder toward the engine: power degrades to 17.85 % with a band of 134.98 kW, and the capacity base contracts to 0.294. Doubling τ_t shrinks less: power improves to 13.85 %, closer to the linear limit of

TABLE IX
PRIOR AGAINST POSTERIOR OF THE 16 PARAMETERS OF CRUSHERBAYES, FITTED TO ALL TEN HP500 SURVEYS; THE RATIO IS THE POSTERIOR STANDARD DEVIATION OVER THE PRIOR STANDARD DEVIATION ON THE NATURAL SCALE, AND THE READING FOLLOWS THE THRESHOLDS 0.5 AND 0.9.

Parameter	Unit	Nominal (prior median)	Prior mean	Prior SD	Posterior mean	Posterior SD	Ratio	Reading
Q_{ref}	t/h	647.00	667.13	167.01	646.35	75.74	0.454	identified
W_i	kWh/t	9.00	9.11	1.52	8.39	1.29	0.854	partly identified
ζ	–	1.30	1.33	0.30	1.14	0.22	0.733	partly identified
P_n	kW	110.00	111.83	47.85	86.35	38.04	0.795	partly identified
s_{K_1}	–	1.00	1.01	0.15	1.07	0.15	0.965	not contracted
s_{K_2}	–	1.00	1.01	0.15	1.05	0.16	1.026	not contracted
K_3	–	2.30	2.31	0.49	2.23	0.49	0.992	not contracted
Δt_{10}	points	0.00	0.00	3.99	−0.11	3.95	0.990	not contracted
σ_{tph}	t/h	60.00	98.92	129.67	145.23	39.04	0.301	identified
σ_{kW}	kW	15.00	24.73	32.42	38.81	10.75	0.332	identified
$d_{\mathrm{tph},0}$	t/h	0.00	0.00	140.00	53.36	122.88	0.878	partly identified
$d_{\mathrm{tph},1}$ (setting)	t/h	0.00	0.00	140.00	100.84	49.00	0.350	identified
$d_{\mathrm{tph},2}$ (f_{80})	t/h	0.00	0.00	140.00	56.57	47.63	0.340	identified
$d_{\mathrm{kW},0}$	kW	0.00	0.00	36.00	−12.57	32.77	0.910	not contracted
$d_{\mathrm{kW},1}$ (setting)	kW	0.00	0.00	36.00	−57.07	13.70	0.381	identified
$d_{\mathrm{kW},2}$ (f_{80})	kW	0.00	0.00	36.00	−5.16	13.03	0.362	identified

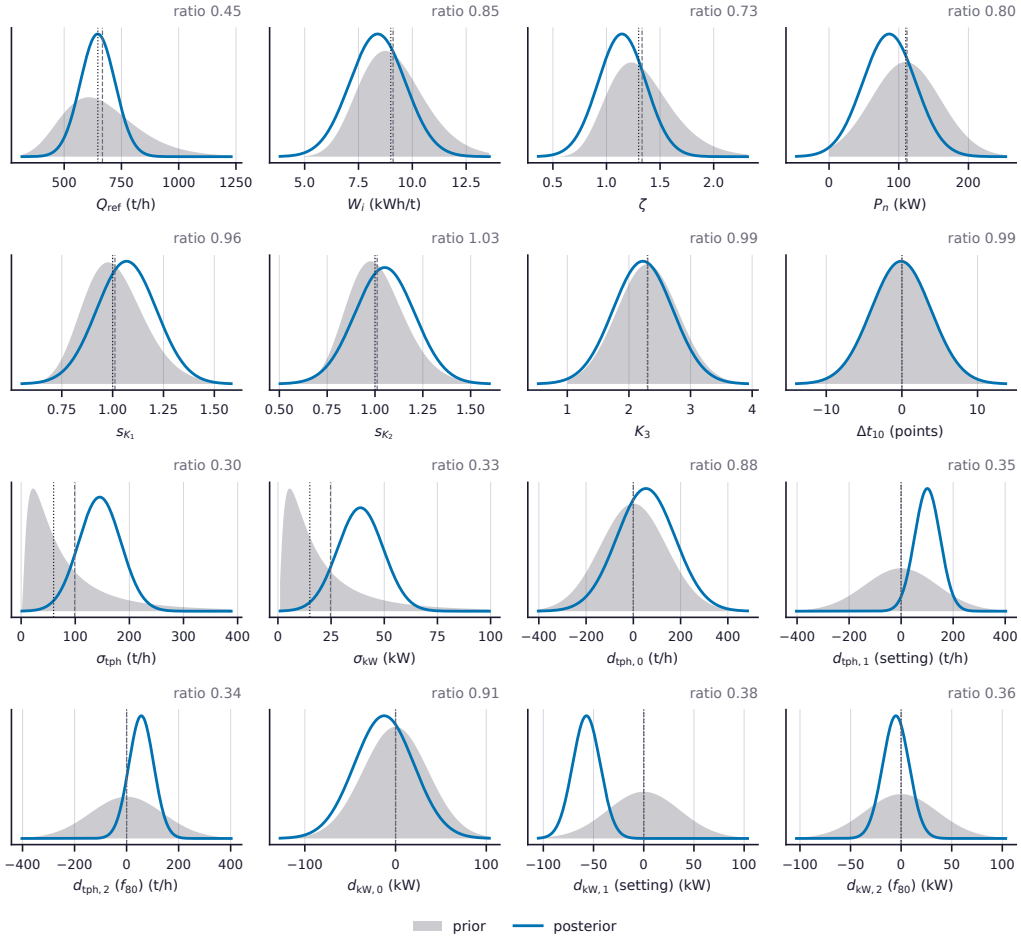


Fig. 4. Prior (grey area) against posterior (blue line) of each of the sixteen parameters, fitted to all ten surveys; the posterior is drawn as a normal density with its mean and standard deviation, the dotted line is the nominal value (the prior median), the dashed line the prior mean, and the ratio is the posterior over the prior standard deviation.

Corollary 1, and the capacity base loosens to 0.638, as (18) states. Throughput moves by at most 0.12 points in these variants and with the doubled prior widths. None of the variants changes the outcome of the dropping rule.

The sampler changes the diagnostics and not the scores. The stretch move gives 12.00 % and 14.29 %, equal to the

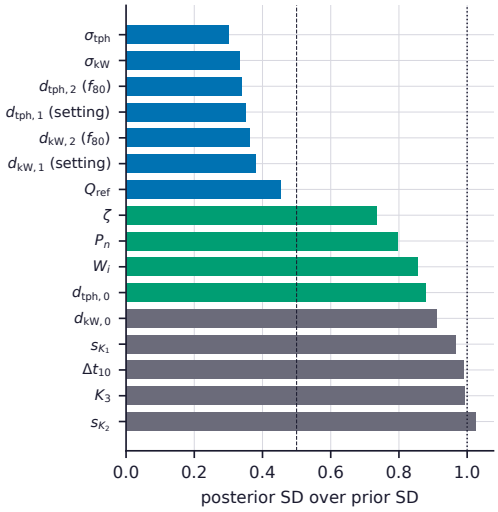


Fig. 5. Posterior over prior standard deviation of each parameter on the fit to all ten surveys: identified (blue, up to 0.5), partly identified (green, up to 0.9) and not contracted (grey), with dashed and dotted lines at 0.5 and 1.

TABLE X
CONVERGENCE OF THE 11 POSTERIORS OF THE METHOD: THE LARGEST INTEGRATED AUTOCORRELATION TIME $\tau_{\max}$ OVER THE TEN DIMENSIONS, THE POST-BURN CHAIN LENGTH IN UNITS OF $\tau_{\max}$ (THE CRITERION IS 50), THE MEAN ACCEPTANCE FRACTION AND THE RETAINED SAMPLES.

Posterior	$\tau_{\max}$ (steps)	Post-burn steps / $\tau_{\max}$	Acceptance	Retained samples
without survey 1	52.6	114.1	0.204	12000
without survey 2	52.7	113.8	0.201	12000
without survey 3	54.7	109.7	0.203	12000
without survey 4	84.8	70.7	0.171	12000
without survey 5	52.7	113.9	0.203	12000
without survey 6	48.0	125.0	0.204	12000
without survey 7	55.1	108.9	0.200	12000
without survey 8	48.9	122.7	0.204	12000
without survey 9	52.6	114.1	0.203	12000
without survey 10	50.7	118.3	0.202	12000
all ten surveys	48.6	123.5	0.204	12000

differential-evolution moves within the seed spread, but none of the eleven posteriors meets the criterion: its autocorrelation time is 126.4 to 200.5 steps, so the post-burn chain is below 50 autocorrelation times in every posterior (Fig. 7). The choice of the move is a choice of reliability of the posterior summaries, which are the identifiability result.

Figure 8 draws the held-out scores of every variant against the seed spread of the full method. It does not show what the variants do to identification (the contractions quoted above) nor their convergence, which Table XI gives, and its scores carry no bootstrap interval, so a variant is read against the seeds and not against the survey-level uncertainty. What follows from it is that on power every variant but the stretch move shifts the score by more than the seed spread, and that the removal of the discrepancy and the halving of its scale shift it most.

IX. DISCUSSION

A. Why the point accuracy does not improve

Proposition 3 answers it. Given the scalars, the predictive mean is a ridge regression of the survey on the standardized setting and feed size, shrunk with a weight of about one tenth toward the engine's best plane, plus the engine's non-affine remainder. On throughput the engine's plane is the engine itself (Proposition 2), and the method is a blend of the free-form fit (12.18 %) and the refit capacity line (12.09 %); it ties both, as the paired differences show. On power the engine's plane has the wrong sign of the setting slope, so shrinking toward it costs accuracy, and the closed form traces the cost as λ grows (Fig. 6); the remainder $e_{t,h}$ is small because most of the engine's power variation is affine in the two regressors. A line in the setting alone already captures the strongest dependence of the data, and with ten surveys a second regressor costs more in variance than it gains. The negative result on power is therefore structural, not a failure of the sampler, and it would persist with any amount of computation on these data.

B. Why the population balance is only weakly identified

The two measured outputs carry the product size only through the Bond term of power, and the Bond draw of the engine spans only 196 to 237 kW across surveys whose power spans 187 to 355 kW. More than nine tenths of the power sensitivity of each population-balance scalar is an affine function of the setting and the feed size, which the discrepancy discounts by a factor of about one tenth (Proposition 1). What is left is too little to narrow the four scalars, which keep the width of their prior, but enough to move the two classification multipliers by about a third of a prior standard deviation and to change the held-out power MAPE by 0.23 points when they are fixed, both beyond the Monte Carlo error and both far inside the survey-level uncertainty (Remark 1). The engine's product size is therefore not validated by these surveys, and the direction in which they move the classification is not evidence about the machine, since it is driven by a power target that the source estimated from electric current. Measured product size distributions per survey would constrain them directly; the distributions of the source are figures only, so this was not tested. Proposition 1 also says how much such data would help: a measured size has a sensitivity to the classification constants that is not an affine function of the setting and the feed size, so it would be counted at the full weight $1/\sigma^2$ rather than at the discount of the plane.

C. Confounding of calibration and discrepancy

Brynjarsdóttir and O'Hagan [30] show that the discrepancy is confounded with the calibration parameters. Here the confounding has an exact form. The capacity base is identified only through the finite width of the throughput discrepancy prior, and its contraction moves with τ_{tph} from 0.294 to 0.638 when the scale is halved or doubled. The no-load power is a constant, confounded with the constant of the power discrepancy, and both are only partly identified or not contracted. The work index and the power slope enter as a product, and each of them is only partly identified. The posterior of an individual scalar should therefore not be read as an estimate of a physical value of the ore or the machine: W_i at 8.39 $\pm$ 1.29 kWh/t is the prior reweighted along a ridge, not a measurement of the work index of the itabirite. The

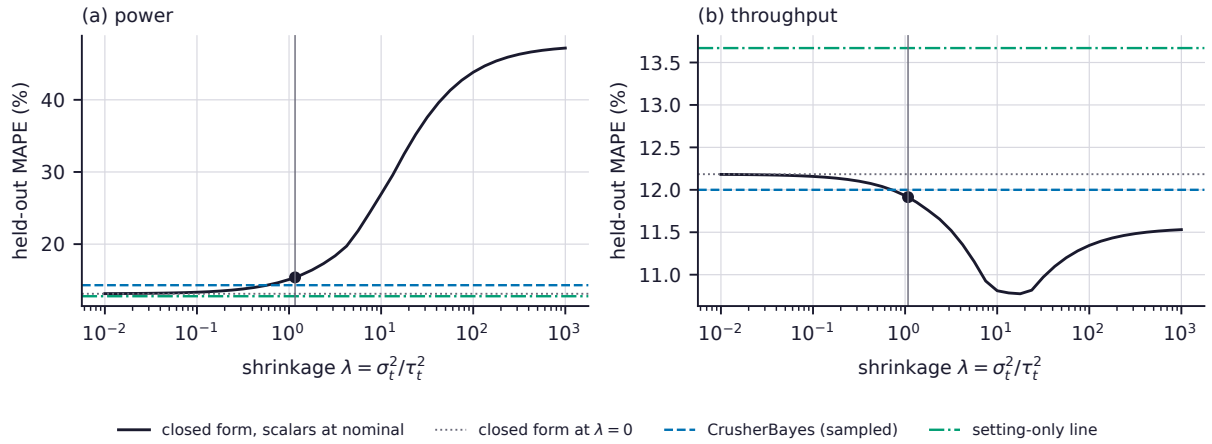


Fig. 6. Held-out MAPE of the closed-form predictive mean (19) with the engine scalars at their nominal value, against the shrinkage λ, for (a) power and (b) throughput; the dot marks the λ of the method, the dashed line the sampled CrusherBayes and the dash-dotted line the setting-only line.

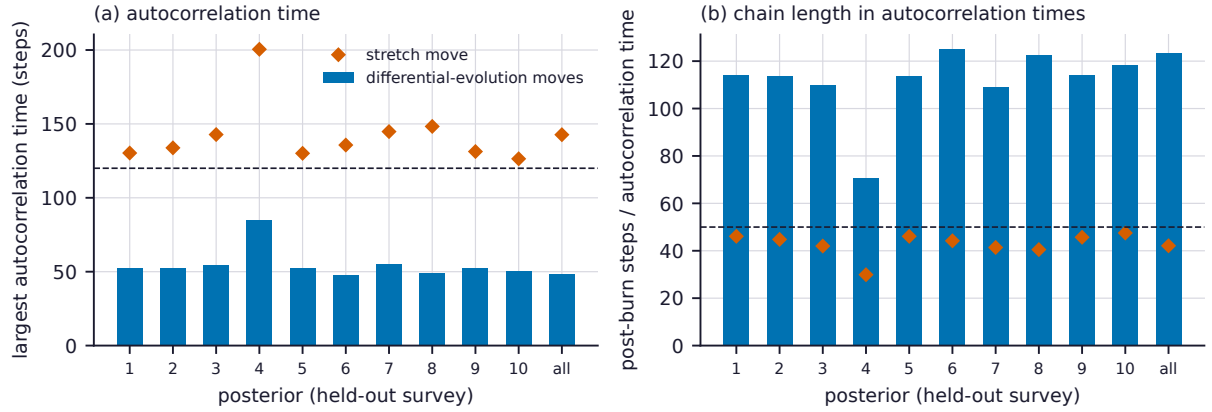


Fig. 7. Convergence of the eleven posteriors (ten folds and all ten surveys): (a) the largest integrated autocorrelation time and (b) the post-burn chain length in autocorrelation times, for the differential-evolution moves (bars) and the stretch move (diamonds); the dashed lines mark the criterion, an autocorrelation time of at most 120 steps in (a) and a chain of at least 50 autocorrelation times in (b).

TABLE XI
ABLATION OF CRUSHERBAYES ON THE TEN HP500 SURVEYS: EVERY ROW RERUNS THE ELEVEN POSTERIORS OF THE STRICT LEAVE-ONE-SURVEY-OUT PROTOCOL WITH ONE COMPONENT CHANGED, AND GIVES THE HELD-OUT MAPE, COVERAGE AND MEAN WIDTH OF EACH OUTPUT, THE RANGE OF THE LARGEST AUTOCORRELATION TIME AND THE CONVERGED POSTERIORS.

Variant	Throughput			Power			$\tau_{\max}$	Converged
	MAPE	Cov.	Width	MAPE	Cov.	Width	(steps)	
The method (seed 42)	12.00	0.8	439	14.31	0.9	117.7	48–85	11 of 11
The method, seed 43	11.99	0.8	438	14.29	0.9	117.8	50–80	11 of 11
The method, seed 44	11.99	0.8	439	14.31	0.9	117.7	47–73	11 of 11
Without the discrepancy	11.98	0.7	451	27.15	0.7	190.2	54–76	11 of 11
Engine scalars fixed at nominal	11.95	0.8	425	15.99	0.7	124.5	34–39	11 of 11
Population-balance scalars fixed	11.99	0.8	439	14.08	0.8	117.7	46–54	11 of 11
Engine prior widths doubled	12.08	0.8	440	14.43	0.9	119.2	60–142	10 of 11
Discrepancy scale τ_t halved	11.89	0.9	435	17.85	0.8	135.0	49–108	11 of 11
Discrepancy scale τ_t doubled	12.12	0.8	446	13.85	0.8	117.3	47–89	11 of 11
Stretch move instead of differential evolution	12.00	0.8	440	14.29	0.9	117.8	126–200	0 of 11

same holds for ζ and P_n, whose posterior means of 1.14 and 86.35 kW are the values that, together with the discrepancy, reproduce ten power measurements.

D. What the band is worth, and the negative results

The band of CrusherBayes comes from a generative model of a survey (engine, discrepancy and noise), so it carries the

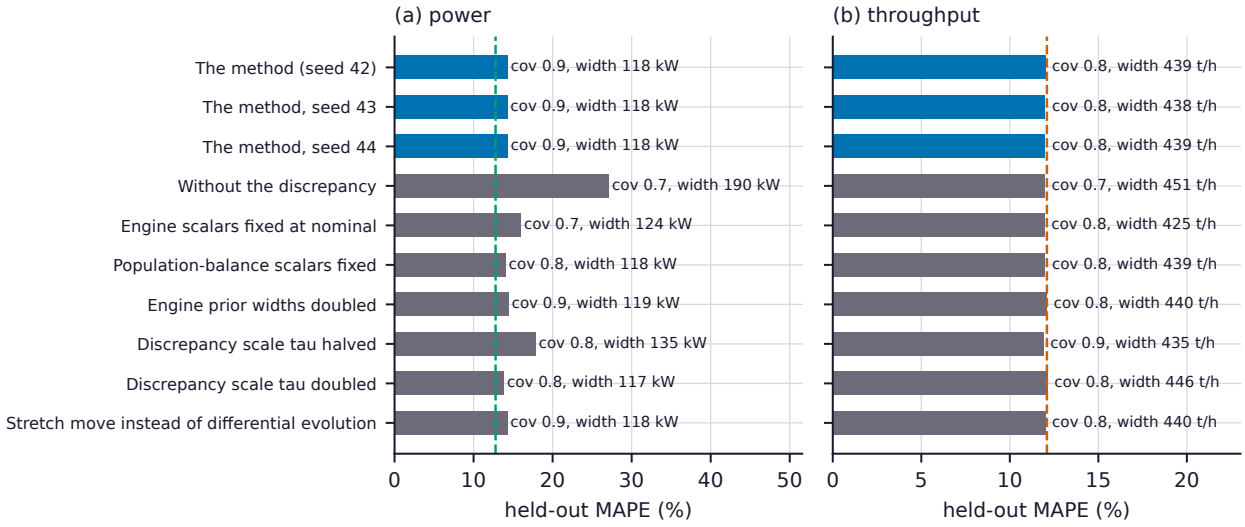


Fig. 8. Held-out MAPE of each ablation variant for (a) power and (b) throughput, with its coverage and mean band width as labels; the dashed line is the setting-only line for power and the refit capacity base for throughput, and the three seeds of the full method are in blue.

uncertainty of the scalars and of the discrepancy, and its width is set by the posterior of the noise. Against the residual band of the refit engine it gives the same coverage at 43.9 % less width on power, a gain that is plain in the data. Against the setting-only line it gives about the same width with one more survey covered out of ten, which the Wilson intervals do not resolve. Three results are negative and are part of the finding: the method does not beat a line in the setting on power; the engine-based power models are no better than the constant mean, with the refit power form (1.47 points against the mean, paired interval −6.23 to 12.00), its bounded residual (0.89, −14.89 to 19.54) and the Bond draw (2.40, −12.55 to 15.87) within the noise, and Morrell through the origin (13.61, 6.50 to 22.91) and the engine at its shipped constants (23.10, 7.16 to 39.87) clearly worse; and the refit power form on all ten surveys takes a negative slope (−2.41, with an intercept of 828.6 kW), which is the engine's power moving against the measured power. The Morrell form with a no-load term reaches 15.21 %, but its fitted work index is negative in every fold, so it is a descriptive line in the engine's size term and not an estimate of M_{ic}; through the origin its fitted M_{ic} is 3.70 to 4.21 kWh/t across folds and its error is 38.08 %. For a plant engineer the practical reading is that, on these surveys, the population balance engine contributes a throughput line as accurate as the simple fits and a power band narrower than its own residual band, and the power prediction comes from the data's own dependence on the setting.

X. Limitations

A. Internal validity

The priors and the discrepancy scales are choices: we found no published range of the fitted scalars for this ore. The ablation measures their effect (doubling the prior widths, halving and doubling τ_t) and finds that accuracy moves by at most 3.54 points on power and 0.12 points on throughput, while the identification of the capacity base moves with τ_{tph} as the theory states; a different prior family was not tried. The prior median of each noise scale and the scale τ_t of each discrepancy were set at about 5 and 12 % of typical levels of 1,200 t/h and 300 kW read from all ten surveys, so every fold's prior carries a little information from its held-out survey; the doubling and halving of τ_t bound its effect on the scores. The discrepancy prior of Kennedy and O'Hagan was kept; the orthogonal discrepancy prior of Plumlee [33], which removes the confounding of Proposition 2, was not tried. The convergence criterion uses the autocorrelation time of one ensemble per posterior, and the Monte Carlo error is measured by three seeds, not by independent ensembles started from dispersed points. The linearized posterior of Section V is evaluated at one point (the posterior mean on all ten surveys) and ignores the truncation of the priors; it differs from the sampler by 0.096 on P_n, for reasons that were not separated. The comparators that use the engine's P_{80} (Morrell, Bond) inherit the engine's product size, which is not measured. The engine evaluates its classification, breakage and Bond draw at its grid F_{80}, which on the two coarsest feeds lies below the measured f_{80} (118.9 against 140 mm on survey 7, 109.9 against 123 mm on survey 8), while the discrepancy uses the measured value. The seed of the bootstrap is fixed, so its intervals are reproducible.

B. External validity and construct

The power target is a construct threat to every power result. The source estimated each power value from electric current, and its authors relate their own poor fit on power to the measurement of power on crushers operating in parallel [8]. The main negative result (H1 on power), the identification of ζ, W_i and P_n, and the location shift of the classification multipliers all rest on that target. A per-survey measurement error of this origin enters the model as noise or as discrepancy, so σ_{kW} and the power discrepancy may describe the power

estimate rather than the crusher, and a metered motor power could change both the accuracy ranking and the identification on power; the throughput results do not depend on it.

Ten surveys of one machine, one ore and one operation, with two measured outputs and two regressors, bound every result. Coverage is coarse: one survey moves it by 0.1, and the Wilson intervals of the coverages span 0.49 to 0.98. The throw and the speed are fixed at the machine's nominal values, so Proposition 2 holds exactly here and would not hold with a varying speed; the feed gradation is rebuilt from the measured f_{80} with a Rosin–Rammler distribution of fixed uniformity, so neither the throw, the speed nor the feed shape is calibrated. The discrepancy is linear in two regressors with a fixed prior scale, not a Gaussian process with estimated hyperparameters, and the two outputs are independent given the scalars, although the measured power and throughput are correlated at -0.85. Whether the band stays calibrated at a setting outside the surveyed range of 38 to 55 mm was not tested. The MAPE weights a survey by the inverse of its measured value, so the low-power surveys weigh more on power. The comparison is limited to methods that can be run from published numbers: the throughput model of Rocha et al. is run, with a work index of the blend and a moisture that the source does not print, but their drop-weight power model is not, because the source prints neither the size distribution of each survey nor the energy at the t_{10} of most surveys; discrete-element simulation is not covered, and the literature search was not exhaustive. The rule for dropping the method is lenient by construction, since it requires failure on accuracy and on calibration at once.

XI. Reproducibility

The code, data and artifacts are in the repository `github.com/fsantibanezleal/CAOS_ChancaDEM`, release 0.13.000 of the product. The posteriors of the method are sampled by an offline pipeline step (`python data-pipeline/run.py hp500`), which writes `data/derived/hp500/crusherbayes.json`; the engine, the priors, the likelihood, the sampler and the scores are in `data-pipeline/pipeline/hp500/crusherbayes.py`, and the comparators in `data-pipeline/pipeline/hp500/loo.py`. The ablation and the seed replicates are run by `python manuscripts/crusherbayes/ablation/run_ablation.py`, which calls the same code and writes one document per variant under `manuscripts/crusherbayes/data/ablation/`; each run took 19 to 47 minutes on a workstation with 32 logical processors, the eleven posteriors sharing a pool of worker processes (the documents of these runs record the processor count, not the number of workers). The checks of the theory are computed by `python manuscripts/crusherbayes/analysis/theory_check.py` into `manuscripts/crusherbayes/data/theory.json`, and the wall time of one engine evaluation by `python manuscripts/crusherbayes/analysis/engine_timing.py` into `manuscripts/crusherbayes/data/engine_timing.json`, on the same workstation. Every figure, table and number of the text is regenerated from these artifacts by `python manuscripts/crusherbayes/figures/make_figs.py`, which writes the number-to-source table `manuscripts/crusherbayes/data/claims.json`. The sampler is `emcee` 3.1.6 with seed 42; the bootstrap uses seed 42. The document of the method carries the SHA-256 of its inputs,

```
483d24c5ddd35b41f2be05628fe9f09fd22562c5ab59cda69e036b9bb0ce605a
```

and the SHA-256 of the document read here is

```
ce57ab2db4727dd73a780d534293938e575ca88454e5dc98d8042d2dcaec1316
```

XII. Conclusions

Ten HP500 surveys, two measured outputs and a linear structural discrepancy identify the noise and the dependence of both outputs on the setting and the feed size, and the capacity base only through the finite scale of the discrepancy prior, with a contraction of 0.294 to 0.638 as that scale is halved or doubled. They determine the work index and the power slope only as a product. The classification and breakage scalars of the population balance keep the width of their prior (ratios 0.965 to 1.026), which the theory predicts and a linearized posterior reproduces, compared on the natural scale, within 0.024 on thirteen of fourteen parameters. The conjecture that the surveys leave them exactly at their prior is refuted: the two classification multipliers move by up to 0.37 prior standard deviations in all three seeds, and fixing the four scalars changes the held-out power MAPE by 0.23 points against a seed spread of 0.02. Held out, CrusherBayes reaches 12.00 % on throughput and 14.31 % on power, no better than a line in the closed-side setting (12.78 % on power, paired difference 1.54 points, interval -1.11 to 4.10), and on throughput it ties the published model of Rocha et al. (paired difference 1.38 points, -5.81 to 7.93), whose power model the published numbers do not allow to run. Its 80 % bands cover 8 and 9 of the ten held-out surveys and are 43.9 % narrower on power than the residual band of the refit engine; ten surveys cannot tell these coverages from those of a slightly miscalibrated band. The ablation shows that the discrepancy carries the power prediction and that the population-balance scalars change the scores by far less than the survey-level uncertainty. Every power result rests on a power that the source estimated from electric current. To calibrate the population balance itself, the surveys need to measure the product size and a metered power; the next steps are the digitized product size distributions of the source with their digitization error, and surveys at a varying speed, which would break the exact confounding of the capacity base.

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